



Discrete Optimization

An energy-efficient scheduling approach for a two-stage hybrid flow shop with parallel batch machines

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ABSTRACT

With the rise of energy costs and the deterioration of the environment, manufacturing companies face increasing production expenses. Reducing energy consumption while improving manufacturing efficiency has become a major challenge. This study addresses the scheduling problem of a **two-stage hybrid flow shop** with **time-of-use (TOU) electricity pricing**, which arises from a tempered glass production environment. In the first stage, **parallel machines** are subject to specific qualification requirements, while **parallel batch processing machines** are used in the second stage. The objectives are to minimize both the makespan (C_{max}) and total energy consumption (TEC). To address this problem, a mixed-integer programming (MIP) model is developed. Furthermore, a **constructive heuristic** based on greedy local search and an **adaptive multi-population cooperative evolutionary algorithm (AMCEA)** are proposed, respectively. In AMCEA, subpopulations are formed based on reference vectors, and adaptive cooperative search facilitates information exchange. Two regional local search operators and two adjustment strategies based on TOU electricity pricing are designed to enhance the exploitation capability of the proposed algorithm. Experimental results demonstrate that the proposed algorithm outperforms state-of-the-art algorithms for the hybrid flow shop under TOU electricity pricing problem.

1. Introduction

With sustained economic growth and population increase, the global energy demand in the manufacturing industry has been increasing. Statistical data show that the manufacturing industry accounts for nearly one-third of global energy consumption. In major manufacturing countries such as China, the energy consumption of the manufacturing industry exceeds half of the nation's total energy consumption (Lin & Chen, 2018). This situation has imposed tremendous pressure on global energy resources, exacerbated environmental issues, and posed significant global economic and social stability challenges (Kuzemko et al., 2017). Therefore, realizing the importance of green manufacturing and low-carbon development, an increasing number of enterprise managers have actively begun to explore technologies and strategies to reduce energy usage and pollution while promoting energy conservation and sustainable production (Jiang et al., 2010; Omer, 2008).

The tempered glass manufacturing industry is characterized by high energy consumption and emissions, particularly in the two key

stages of glass production and tempering, where energy use and waste generation are relatively high. Therefore, this research takes tempered glass production as the starting point and divides the process into two stages. The first stage is glass processing, which requires the qualification requirements of the machines. The second stage is tempering, which involves multiple parallel large-batch furnaces and is subject to energy consumption and capacity constraints. These two stages constitute a scheduling problem for a **hybrid flow shop under time-of-use (TOU) electricity pricing (HFS-TOU)**. Through modeling and optimization of the scheduling problem, this study aims to achieve the bi-objectives of shortening production time and reducing energy consumption (i.e., lowering electricity costs) while accounting for differences in energy usage across different machine states.

Research on the energy-efficient scheduling problem of the hybrid flow shop in tempered glass production is crucial to improving production efficiency, reducing energy consumption and pollutant emissions, and promoting green and sustainable manufacturing. Moreover, it will

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facilitate the ecological transformation of related sectors, such as construction and furniture, thereby making a significant contribution to alleviating the ongoing energy and environmental crises.

The main contributions are summarized as follows:

- A **mixed-integer programming** (MIP) model is developed to address the studied HFS-TOU problem, considering multiple batch processing machines, machine qualification, machine states, different energy consumption rates, and TOU electricity pricing.
- A **constructive heuristic (CH) algorithm** is proposed, which integrates multiple scheduling rules and energy minimization strategies and adopts a greedy strategy to optimize machine scheduling and job processing order. Furthermore, a neighborhood search algorithm is designed to improve solution quality further.
- An **adaptive multi-population co-evolutionary algorithm (AM-CEA)** is proposed, which facilitates information exchange among subpopulations through an adaptive cooperative search strategy and combines regional local optimization and price-based adjustment strategies to improve solution quality and optimization performance.

The remainder of this paper is organized as follows. Section 2 reviews related literature. Section 3 defines the investigated problem. Section 4 describes the proposed algorithm in detail. Section 5 presents the experimental results and comparisons. Section 6 concludes this study and discusses future research directions.

2. Literature review

The tempered glass manufacturing process investigated in this study can be formulated as a two-stage hybrid flow shop (HFS) scheduling problem, which has attracted considerable research interest in recent years. Arthanary (1971) conducted pioneering work on the two-stage flow shop scheduling problem. Subsequently, following this seminal work, researchers have extended their investigations to HFS systems that consider additional real-world constraints such as machine qualification, batch processing, and constraints related to TOU electricity pricing, which reflect the complex features of real-world tempered glass manufacturing (McMaster, 1989). This review highlights studies that address problem constraints similar to those considered in this research. For a more comprehensive literature review of HFS, readers can refer to surveys by Neufeld et al. (2023), Ruiz and Vázquez-Rodríguez (2010), and others.

There are indeed numerous studies on HFS scheduling that consider machine qualification constraints. Elmi and Topaloglu (2013) investigated HFS with unrelated parallel machines in each stage and developed two simulated annealing (SA) algorithms to minimize the makespan. In the study considering additional scheduling constraints, Perez-Gonzalez et al. (2019) designed a constructive heuristic algorithm to minimize total tardiness for unrelated parallel machines with qualification constraints and sequence-dependent setup times. Copado Méndez et al. (2022) proposed a simulated optimization algorithm for HFS with time dependencies and priority constraints to minimize the makespan. Fekri et al. (2024) investigated the flexible flow shop scheduling problem with multi-skill resource constraints, constructed a mixed integer linear programming (MILP) model, and used a genetic algorithm and simulated annealing algorithm to solve it to optimize the total completion time and the total idle time. In addition, Zhang and Chen (2018) developed a novel decoder to handle qualification constraints, which is embedded into a discrete differential evolutionary algorithm (DEA) to solve the reentrant HFS problem. Mlekusch and Hartl (2025) studied a reentrant flexible flow shop scheduling problem with dual resource constraints and proposed a constraint programming model and a hybrid genetic algorithm to minimize the makespan.

Since batch processing machines can process multiple jobs simultaneously, researchers have also paid increasing attention to batch

production environments in HFS scheduling. Wang et al. (2012) established a MIP model for specialized two-stage HFS multiple batch processing machines in semiconductor manufacturing and designed a local search heuristic algorithm to minimize the makespan. Zhang et al. (2017) proposed a hybrid DEA to minimize total completion time for flow shop scheduling with batch and discrete processors with incompatible job families and limited buffers. Considering the different deterioration degrees and sizes of jobs, Liu et al. (2019) combined the distribution estimation algorithm with DEA to propose a hybrid algorithm to optimize makespan. Wu and Cao (2022) studied reentrant HFS with parallel batch processing machines and presented an improved decomposition-based multi-objective evolutionary algorithm to optimize makespan and energy consumption. In addition, Luo et al. (2024) proposed a scheduling method based on deep reinforcement learning to minimize the makespan for the HFS problem with the coexistence of parallel machines and batch processing machines.

In recent years, the energy-related multi-objective HFS scheduling problem has received extensive attention. Researchers have proposed various optimization methods for different scheduling objectives and constraints. Shao et al. (2021) studied a multi-objective distributed HFS problem with the objectives of minimizing makespan, total weighted earliness and tardiness, and total workload and proposed a multi-objective evolutionary algorithm based on multiple neighborhoods local search for the solution. For the energy-aware distributed HFS problem that minimizes the makespan and energy consumption, Wang and Wang (2021) proposed a cooperative memetic algorithm based on reinforcement learning (RL). Wang et al. (2023) investigated minimizing the makespan and total energy consumption in HFS environments where machines have different machining speeds and proposed an NSGA-II algorithm incorporating fuzzy correlation entropy and an energy-aware strategy. Zhang et al. (2024) further addressed the HFS problem with heterogeneous machines and proposed a multi-objective memetic algorithm combining particle swarm optimization (PSO) and Q-learning local search to optimize the makespan and total energy consumption.

In parallel, considering the TOU pricing policies, the energy-related scheduling problem has gained increasing attention. Luo et al. (2013) proposed a new multi-objective ant colony optimization (MOACO) metaheuristic algorithm to solve the HFS problem, considering not only production efficiency but also power costs and the existence of TOU pricing to minimize the planned power costs as much as possible. Li et al. (2018) proposed an energy-aware multi-objective optimization algorithm (EA-MOA) for the HFS scheduling problem with setup energy consumption, which can optimize the minimization of maximum production time and energy consumption while considering energy consumption for setups. In addition, Zhang et al. (2019) investigated the green HFS scheduling problem with different machine processing speeds and designed a multi-objective discrete artificial bee colony algorithm based on the decomposition method to minimize the makespan and total energy consumption. Qin et al. (2022) proposed an energy-efficient iterative greedy algorithm incorporating a swap strategy to solve the blocking HFS problem, aiming to optimize energy consumption and economic costs in production. Furthermore, Liu et al. (2020) abstracted a three-stage HFS problem derived from a real manufacturing system and designed model-based heuristics and a bi-objective DEA to optimize electricity cost and production rate. Similarly, Ho et al. (2021) developed three heuristics based on Johnson's rule and three based on Hadda's algorithms to optimize makespan and electricity cost for two-machine flow shop scheduling with TOU tariffs.

Several studies have been conducted to explore the energy-efficient scheduling problem involving batch processing machines in the tempered glass production process. For example, Wang et al. (2020) investigated a two-phase hybrid flow shop scheduling problem under TOU electricity pricing, where the first stage consists of multiple parallel machines and the second stage involves a single batch processing machine. To address the bi-objective optimization of minimizing makespan and

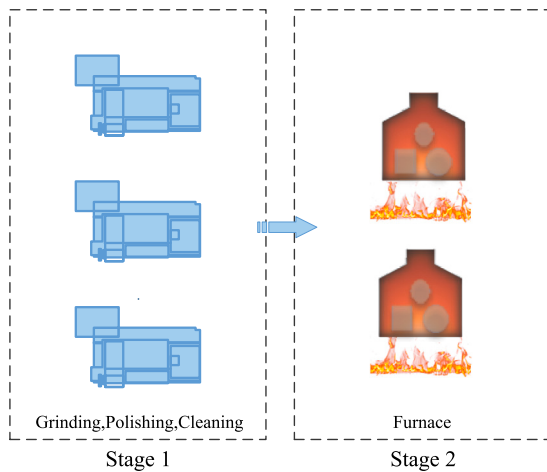


Fig. 1. Production process of tempered glass.

energy consumption simultaneously, a MIP model is developed, and a problem-tailored constructive heuristic, a bi-objective tabu search, and a bi-objective ant colony optimization algorithm are proposed. Although this research provides valuable modeling insights and methods for addressing such problems, its scheduling model exhibits limitations in terms of problem complexity and the realism of energy consumption modeling. In contrast, this paper improves upon previous work in two key aspects. First, the TOU tariff structure is based on actual industrial tariff data from Beijing, enhancing the model's applicability to realistic energy-aware scheduling scenarios. Second, the second stage is extended to include multiple parallel batch processing machines, better reflecting the coordinated operation of such machines in real-world tempered glass workshops. This extension introduces the need for the coordinated scheduling of multiple batch processing machines in the second stage, thereby increasing the complexity of the scheduling problem. To address this challenge, heuristic and metaheuristic algorithms are developed to solve this bi-objective energy-aware scheduling problem. Moreover, to the best of our knowledge, this specific problem has not yet been addressed in the existing literature.

3. Problem formulation

In this section, Section 3.1 explains the studied scheduling problem, and Section 3.2 formulates a MIP model for the investigated problem.

3.1. Problem description

In the manufacturing process of tempered glass, ordinary glass is heated above 600°C, higher than its softening point, and then rapidly cooled. This process creates surface compressive stress and internal tensile stress in the glass, enhancing its strength and impact resistance. The production process is illustrated in Fig. 1.

The processing of glass (jobs) can be divided into two stages based on customer orders for scheduling purposes. In stage 1, jobs of different sizes undergo operations, such as grinding and polishing on parallel machines. These parallel machines are subject to qualification constraints, as not all can process customized shapes. In stage 2, the jobs processed in the first stage are transferred to multiple high-energy furnaces (regarded as batch processing machines) for further processing. These batch processing machines have identical capacities and can process multiple jobs simultaneously up to their capacity. This setup aims to simulate the production scenario closely while simplifying the modeling and solving the problem.

The assumptions on the studied problem are summarized as follows.

- (1) All jobs are available at the start, i.e., their release times are zero.
- (2) Setup times are negligible compared to processing times.
- (3) Jobs are scheduled non-preemptively on any machine.
- (4) Each job is first processed on a parallel machine in stage 1, then on a batch processing machine in stage 2.
- (5) Machines are always available, and no disruptions or breakdowns are allowed.
- (6) In stage 1, each machine processes at most one job at any time without interruption.
- (7) In stage 2, each batch processing machine can process multiple jobs simultaneously within its capacity.
- (8) The processing time and release time of each batch are determined by the longest processing time and the latest release time among the jobs in the batch, respectively.
- (9) Machines have three states (processing, idle, and turned off), each corresponding to a different energy consumption rate. The “turned off” state indicates that a machine stops accumulating further idle energy consumption after completing its last assigned job. It does not represent a physical shutdown or a low-power mode.
- (10) A machine will not be turned off until all assigned jobs are completed.

According to the three-field notation (Graham et al., 1979), the studied problem can be described as $HFS(m_1, m_2) | TOU, p - batch(II), s_j, p_j, Q | C_{max}, TEC$, where HFS denotes a hybrid flow shop environment; m_1 and m_2 represent the number of machines in stage 1 and stage 2, respectively; TOU refers to time-of-use electricity pricing; $p - batch(II)$ indicates the processing environment in stage 2 consists of batch processing machines; s_j and p_j indicate that the jobs have non-identical sizes and processing times; Q denotes the capacity of a batch processing machine; C_{max} and TEC represent the objectives of minimizing makespan and total energy consumption, respectively.

3.2. Model formulation

This study categorizes the assumptions into two types: intra-stage constraints, such as machine eligibility in stage 1 and batch capacity in stage 2, and inter-stage constraints, where the next stage can only start after the previous stage has been completed. The key coupling relationships are as follows: the completion time of a job in stage 1 directly determines the earliest start time of its corresponding batch in stage 2, and batch grouping is subject to the capacities of batch processing machines and the sizes of jobs. The symbols used are defined in Tables 1 and 2 to facilitate the model formulation. Based on these definitions, a mixed-integer programming (MIP) model is developed in this section.

The mathematical formulation for the studied problem is as follows (see Box I and Box II).

The optimization objectives are to simultaneously minimize the makespan and total energy consumption, represented by Eq. (1). Constraints (2) to (14) describe the constraints for stage 1. Specifically, Constraint (2) ensures that each job is processed on only one machine in stage 1. Constraint (3) ensures machine eligibility: if job j cannot be processed on machine i , it cannot be assigned to that machine. Constraints (4) and (5) indicate that if job j is not assigned to machine i , it will not be processed on machine i during any period t . If job j is assigned to machine i , there will be at least one time period t when machine i processes job j . Constraint (6) requires that if job j is processed on machine i in period t , its processing time in stage 1 must equal p_j^1 . Constraints (7) and (8) prohibit the preemption of job j during processing. Constraint (9) ensures machine i is in only one of three states in each period t , i.e., processing, idle, or turn-off. Constraint (10) defines the processing status of machine i in period t in stage 1. Constraint (11) defines the idle status of machine i in period t in stage

Table 1
Indices, parameters and sets.

Notations	Definitions
Indices:	
j, j'	The index of a job, $j, j' \in \{1, 2, \dots, n\}$.
g	The index of a stage, $g \in \{1, 2\}$.
i, i'	The machine index in stage g , $i, i' \in \{1, 2, \dots, m_g\}$.
b	The index of a batch, $b \in \{1, 2, \dots, \alpha\}$.
t, t'	The index of time period, $t, t' \in \{1, 2, \dots, \gamma\}$.
Parameters and sets:	
n	The number of jobs.
N	The set of jobs, $N = \{1, 2, \dots, n\}$.
G	The set of stages, $G = \{1, 2\}$.
m_g	The number of machines in stage g .
M_g	The set of machines in stage g , $M_g = \{1, 2, \dots, m_g\}$.
α	The number of batches ; originally, $\alpha = n$.
B	The set of batches, $B = \{1, 2, \dots, \alpha\}$.
γ	The number of periods .
T	The set of time periods, $T = \{1, 2, \dots, \gamma\}$.
p_j^g	The processing time of job j in stage g .
s_j	The size of job j .
Q	The capacity of each batch processing machine.
ep_i^g	Processing energy consumption rate of machine i in stage g .
ed_i^g	Idle energy consumption rate of machine i in stage g .
eo_i^g	Turn-off energy consumption rate of machine i in stage g (eo_i^g is always a zero vector).
$f(t)$	The unit energy price for time period t .
L	A very large positive number.

Table 2
Decision variables.

Notations	Definitions
px_{ijt}	1 if job j is processed on machine i at period t ; 0 otherwise.
π_{ij}	1 if job j can be processed on machine i in stage 1; 0 otherwise.
a_{ij}	1 if job j is assigned on machine i in stage 1; 0 otherwise.
xp_{it}^g	1 if machine i is processing at period t in stage g ; 0 otherwise.
xd_{it}^g	1 if machine i is idle at period t in stage g ; 0 otherwise.
xo_{it}^g	1 if machine i is turn-off at period t in stage g ; 0 otherwise.
ax_{ibj}	1 if job j is assigned to batch b on machine i in stage 2; 0 otherwise.
y_b	1 if batch b includes at least one job; 0 otherwise.
u_{ib}^t	1 if batch b is processed on batch processing machine i during period t ; 0 otherwise.
c_j^g	The completion time of job j in stage g .
r_{ib}^2	The start time of batch b on machine i in stage 2.
pt_{ib}^2	The processing time of batch b on machine i in stage 2.
ct_{ib}^2	The completion time of batch b on machine i in stage 2.
C_{max}	The makespan of the entire process.
TEC	The total electricity cost consumed in the entire process.
β_{it}^g	Binary auxiliary variables that coordinate the state of use before and after machine i is shut down. (1 if machine i is shut down at period t in phase g and machine i has been used before that time; 0 otherwise.)

1: if machine i is idle in period t , it must process for at least one period from period $t + 1$ to γ ; otherwise, machine i is turned off. Constraint (12) prohibit all machines in stage 1 from being idle in the last periods. Constraints (13) and (15) define the turn-off status of machine i in period t in stage 1: if being turned off in period t , it cannot process any jobs from period $t + 1$ to γ .

Constraints (16) to (36) describe the constraints for stage 2. Specifically, Constraint (16) requires each job to be assigned to only one batch processing machine in stage 2. Constraint (17) ensures that the batch size does not exceed the capacity Q of the batch processing machine. Constraint (18) sets the batch processing time to zero if there is no job in the batch. Constraints (19) and (20) determine whether batch b is empty. If batch b is not empty, the number of jobs in batch b exceeds y_b ; if batch b is empty, no job is assigned to it. Constraint (21) ensures that machine i processes one batch in period t in stage 2 if it is in the processing state. Constraint (22) defines the processing time of each batch. Constraint (23) defines the batch processing time as the longest processing time among the jobs in that batch. Constraint (24) ensures the processing of batch b in stage 2 cannot start until all jobs in batch b have been completed in stage 1. Constraint (25) ensures that the start time of batch $b + 1$ on machine i cannot be earlier than the completion of batch b on machine i . Constraint (26) defines the

completion time of batch b on machine i . Constraints (27) and (28) prohibit interruptions during batch processing on machine i . Constraint (29) ensures machine i is in one of three states in period t . Constraint (30) defines the idle status of machine i in stage 2: if machine i is idle in period t , it must process for at least one period from period $t + 1$ to γ ; otherwise, machine i is turned off. Constraint (31) prohibits all machines in stage 2 from being idle in the last periods. Constraints (32) and (34) define the turn-off status. If turned off in period t , machine i cannot process jobs from period $t + 1$ to γ . Constraint (35) bounds the overall schedule within the predefined time horizon. Constraint (36) defines the maximum completion time. Constraint (37) defines the total energy consumption. Constraints (38) and (39) specify the ranges of variables.

4. Proposed algorithm

A **constructive heuristic (CH)** algorithm based on a **greedy strategy** is proposed in Section 4.1. The algorithm offers advantages such as **fast computation speed**, **high interpretability**, and **broad applicability**. However, due to its reliance on simple combinations of scheduling rules, the algorithm **struggles to obtain the best solution**. Therefore, an **adaptive multi-population cooperative evolutionary algorithm** (AM-

$$\begin{aligned}
 & \min (C_{\max}, TEC) & (1) \\
 \text{s.t. } & \sum_{i=1}^{m_1} a_{ij} = 1 & j \in N & (2) \\
 & a_{ij} \leq \pi_{ij} & j \in N; i \in M_1 & (3) \\
 & a_{ij} \geq px_{ijt} & j \in N; i \in M_1; t \in T & (4) \\
 & a_{ij} \leq \sum_{t=1}^{\gamma} px_{ijt} & j \in N; i \in M_1 & (5) \\
 & \sum_{i=1}^{m_1} \sum_{t=1}^{\gamma} px_{ijt} = p_j^1 & j \in N & (6) \\
 & (1 - \sum_{i=1}^{m_1} px_{ijt})L + t \geq c_j^1 - p_j^1 & j \in N; t \in T & (7) \\
 & (1 - \sum_{i=1}^{m_1} px_{ijt})L + c_j \geq t + 1 \Rightarrow t < c_j & j \in N; t \in T & (8) \\
 & xp_{it}^1 + xd_{it}^1 + xo_{it}^1 = 1 & i \in M_1; t \in T & (9) \\
 & xp_{it}^1 = \sum_{j=1}^n px_{ijt} & i \in M_1; t \in T & (10) \\
 & (1 - xd_{it}^1)L + \sum_{t'=t+1}^{\gamma} xp_{it'}^1 \geq 1 & i \in M_1; t \in T \setminus \{\gamma\} & (11) \\
 & xd_{it}^1 = 0 & i \in M_1 & (12) \\
 & (1 - xo_{it}^1 + \beta_{it}^1)L \geq \sum_{t'=1}^{t-1} xp_{it'}^1 & i \in M_1; t \in T & (13) \\
 & (2 - xo_{it}^1 - \beta_{it}^1)L \geq \sum_{t'=t+1}^{\gamma} xp_{it'}^1 & i \in M_1; t \in T \setminus \{\gamma\} & (14) \\
 & xo_{it}^1 \leq xo_{i,t+1}^1 & i \in M_2; t \in T \setminus \{\gamma\} & (15) \\
 & \sum_{i=1}^{m_2} \sum_{b=1}^{\alpha} ax_{ibj} = 1 & j \in N & (16) \\
 & \sum_{j=1}^n s_j \cdot ax_{ibj} \leq Q \cdot y_b & b \in B; i \in M_2 & (17) \\
 & y_b \cdot L \geq pt_{ib}^2 & b \in B; i \in M_2 & (18) \\
 & \sum_{j=1}^n ax_{ibj} \geq y_b & b \in B; i \in M_2 & (19) \\
 & y_b \geq ax_{ibj} & j \in N; b \in B; i \in M_2 & (20) \\
 & \sum_{b=1}^{\alpha} w_{ib}^t = xp_{it}^2 & i \in M_2; t \in T & (21) \\
 & \sum_{t=1}^{\gamma} w_{ib}^t = pt_{ib}^2 & b \in B; i \in M_2 & (22) \\
 & pt_{ib}^2 \geq p_j^2 \cdot ax_{ibj} & j \in N; b \in B; i \in M_2 & (23) \\
 & (1 - ax_{ibj})L + rt_{ib}^2 \geq c_j^1 & j \in N; b \in B; i \in M_2 & (24) \\
 & rt_{ib}^2 + pt_{ib}^2 \leq rt_{ib+1}^2 & b \in B \setminus \{\alpha\}; i \in M_2 & (25)
 \end{aligned}$$

Box I.

CEA) is further proposed in Section 4.2. As a meta-heuristic algorithm, it can adaptively adjust the search strategy according to the problem characteristics, thus effectively approximating the Pareto frontier. The algorithm possesses global search capabilities and adaptability, which improve its ability to solve complex problems.

4.1. Constructive heuristic algorithm based on greedy strategy

4.1.1. Framework of the proposed algorithm

In the CH algorithm, the processing sequence of jobs in stage 1 is determined according to various scheduling rules. Then, the start time

$$\begin{aligned}
& rt_{ib}^2 + pt_{ib}^2 = ct_{ib}^2 & b \in B; i \in M_2 & (26) \\
& (1 - w_{ib}^t)L + t \geq rt_{ib}^2 & b \in B; i \in M_2; t \in T & (27) \\
& (1 - w_{ib}^t)L + rt_{ib}^2 + pt_{ib}^2 \geq t + 1 & b \in B; i \in M_2; t \in T & (28) \\
& xp_{it}^2 + xd_{it}^2 + xo_{it}^2 = 1 & i \in M_2; t \in T & (29) \\
& (1 - xd_{it}^2)L + \sum_{t'=t+1}^T xp_{it'}^2 \geq 1 & i \in M_2; t \in T \setminus \{\gamma\} & (30) \\
& xd_{it}^2 = 0 & i \in M_2 & (31) \\
& (1 - xo_{it}^2 + \beta_{it}^2)L \geq \sum_{t'=1}^{t-1} xp_{it'}^2 & i \in M_2; t \in T & (32) \\
& (2 - xo_{it}^2 - \beta_{it}^2)L \geq \sum_{t'=t+1}^T xp_{it'}^2 & i \in M_2; t \in T \setminus \{\gamma\} & (33) \\
& xo_{it}^2 \leq xo_{i,t+1}^2 & i \in M_2; t \in T \setminus \{\gamma\} & (34) \\
& \gamma \geq ct_{ib}^2 & i \in M_2 & (35) \\
& C_{\max} \geq ct_{ib}^2 + (y_b - 1)L & b \in B; i \in M_2 & (36) \\
& TEC = \sum_{i=1}^{\gamma} \sum_{g=1}^2 \sum_{t=1}^{m_g} f(t)(xp_{it}^g \cdot ep_i^g + xd_{it}^g \cdot ed_i^g + xo_{it}^g \cdot eo_i^g) & & (37) \\
& px_{ijt}, a_{ij}, xp_{it}^g, xd_{it}^g, xo_{it}^g, ax_{ibj}, y_b, w_{ib}^t, \beta_{it}^g \in \{0, 1\} & j \in N, g \in G, i \in M_g, t \in T & (38) \\
& c_j^1, rt_{ib}^2, pt_{ib}^2, ct_{ib}^2, C_{\max} \geq 0 & j \in N, b \in B & (39)
\end{aligned}$$

Box II.

of each job and the corresponding machine assignment are determined according to the minimum energy consumption (MEC) rule (Wang et al., 2020), which is a dynamic decision-making mechanism that selects the machine and period for each job with the lowest total energy cost. Under machine eligibility constraints, it considers processing energy consumption rates, TOU electricity pricing, and costs of different machine states. Specifically, the rule traverses all available machines and their idle periods to compute total energy costs. The job is then assigned to the option with the minimum overall cost. Afterward, batching and scheduling in stage 2 are carried out according to various scheduling rules. The algorithm proceeds to the next rule if a specific rule fails to generate a feasible solution. Additionally, a neighborhood search strategy is incorporated to improve solution quality by dynamically adjusting start times of jobs in response to current and future TOU electricity prices, so as to reduce overall energy consumption.

Based on this framework, the entire algorithm is divided into the following four main steps:

- (1) Step 1: Schedule jobs on parallel machines in stage 1. Two combined scheduling rules are used: (i) The Johnson rule (Johnson, 1954) in combination with the MEC rule. The Johnson rule determines the sequence of scheduling jobs, while the MEC rule determines the start processing time and machine assignment. (ii) The longest processing time (LPT) rule based on the first-stage processing time p_j^1 , referred to as the $(p_j^1 + \text{LPT})$ rule, is combined with the MEC rule in stage 1. The MEC rule employs a greedy strategy that schedules jobs to avoid high electricity price periods and high energy-consuming machines when the completion time is constrained. At each step, it selects the start time and machine with the lowest local energy consumption, thereby reducing the total energy cost.
- (2) Step 2: Two rules are used for batch formation: (i) First come first served (FCFS) rule in combination with full batch (FB)

rule (Dupont & Dhaenens-Flipo, 2002). (ii) The shortest processing time (SPT) rule based on the second-stage processing time p_j^2 , referred to as the $(p_j^2 + \text{SPT})$ rule, is combined with the FB rule in stage 2. Based on the completion times of jobs in stage 1, one of the above rules is selected to determine the job sequence in stage 2. The jobs are then grouped into batches according to the FB rule. During this process, each batch is filled as fully as possible to maximize the utilization of batch processing machines. A new batch is created if the current batch lacks sufficient capacity to accommodate a job. The job is assigned to the new batch, and the remaining jobs are subsequently added to the batch following the same logic until all jobs are successfully grouped.

- (3) Step 3: Scheduling of batches in stage 2 using the FCFS rule in combination with the MEC rule.
- (4) Step 4: A neighborhood search algorithm is incorporated to enhance the solution quality.

Based on the steps above, the detailed procedure of the CH algorithm is presented in Algorithm 1. The input to the algorithm includes job and machine parameters, and the output is the non-dominated solution set NS . To efficiently arrange the processing of jobs while avoiding overly large or small search intervals, the estimated upper and lower bounds of the total completion time for all jobs at the initial time are set. Specifically, the longest completion time is estimated as $T_{\max} = 2n(\max_{j \in N}(p_j^1) + \max_{j \in N}(p_j^2))$, and the shortest completion time is estimated as $T_{\min} = \sum_{j=1}^n \frac{p_j^1}{m_1 \cdot K} + \sum_{j=1}^n \frac{s_j}{m_2 \cdot Q} \cdot \max_{j \in N}(p_j^2)$. To ensure sufficient processing time for batch processing operations in stage 2, the maximum completion time of stage 1 needs to be constrained. Thus, a set of upper bounds on the completion time in stage 1 is defined as $\text{bound} = \left\{ \frac{T_{\max}}{3}, \frac{T_{\max}}{2}, \frac{2T_{\max}}{3} \right\}$. The parameter b is iteratively updated to correspond to each of the three time boundaries in the set bound . Initially, $T_{\max}$ is used as the constraint for the overall completion time to generate a preliminary schedule and obtain the makespan $C_{\max}$. This $C_{\max}$ value is used as an upper bound of the overall completion

time in subsequent iterations. Only schedules that meet this limit are considered feasible. This mechanism ensures that the search space is progressively narrowed, with the current objective value obtained from the scheduling scheme serving as a reference for further search. The overall procedure of the algorithm is illustrated in Fig. 2.

Algorithm 1 Constructive heuristic algorithm based on greedy strategy

Input: The capacity of each batch processing machine, Q ; The size of job j , s_j ; The processing time of job j in stage g , p_j^g ; **Machine eligibility proportion in stage 1, R .**

Output: Non-dominated solution set NS .

```

1:  $NS \leftarrow \phi$ .
2: Calculate  $T_{max}$  and  $T_{min}$ .
3:  $T \leftarrow T_{max}$ .
4:  $bound \leftarrow \left\{ \frac{T_{max}}{3}, \frac{T_{max}}{2}, \frac{2T_{max}}{3} \right\}$ .
5: for  $b$  in  $bound$  do
6:   while  $T \geq T_{min}$  do
7:     Schedule jobs in stage 1 using Johnson's rule combined with
       the MEC rule.
8:     if All jobs are scheduled to be completed in stage 1 then
9:       Label:  $\triangleright$  Jump label
10:      Group batches using FCFS rules in combination with FB
       rules.
11:      Schedule batches in stage 2 using FCFS rule in combina-
       tion with MEC rule.
12:      if All batches are scheduled to be completed in stage 2
       then
13:        Calculate the objective values  $C_{max}$  and  $TEC$  and
        update  $NS$ .
14:         $T \leftarrow C_{max} - 1$ .
15:      else
16:        Group batches using  $p_j^2 + SPT$  rule in combination
        with FB rule.
17:        Schedule batches in stage 2 using FCFS rule in combina-
        tion with MEC rule.
18:        if All batches are scheduled to be completed in stage
        2 then
19:          Calculate the objective values  $C_{max}$  and  $TEC$  and
          update  $NS$ .
20:           $T \leftarrow C_{max} - 1$ .
21:        else
22:          No feasible solution.
23:        end if
24:      end if
25:    else
26:      Schedule jobs in stage 1 using  $p_j^1 + LPT$  rule in
       combination with MEC rule.
27:      if All jobs are scheduled to be completed in stage 1 then
28:        goto Label.
29:      else
30:        No feasible solution.
31:      end if
32:    end if
33:  end while
34: end for
35: Apply the neighborhood search algorithm to update  $NS$ .
  
```

4.1.2. Neighborhood search algorithm

The basic idea of the neighborhood search algorithm is to consider a sequence of jobs processed consecutively on the same machine, referred to as a job block. The motivation for this approach stems from the impact of TOU electricity pricing policies. The algorithm determines the feasible start time intervals by traversing the job blocks on each machine and calculating their corresponding time windows. The earliest feasible start time is constrained by the completion time of

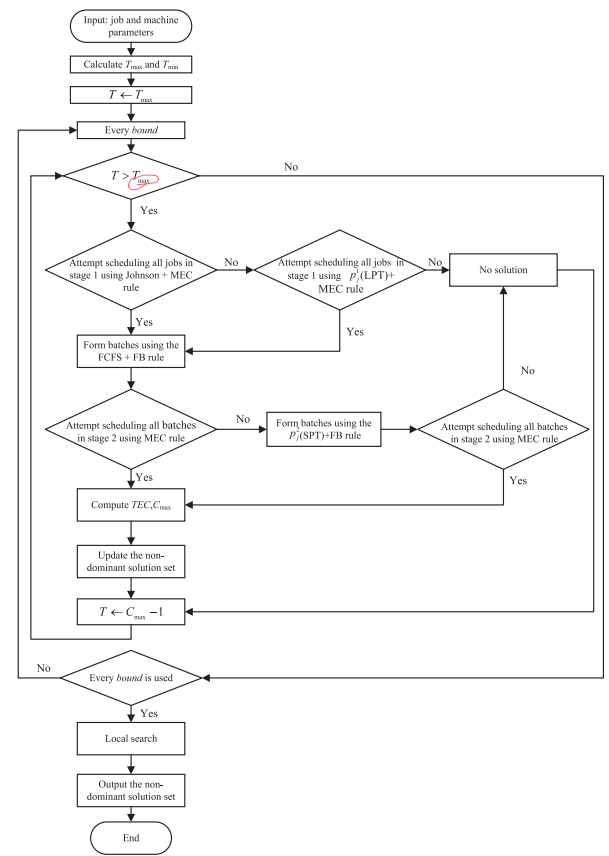


Fig. 2. The flowchart of the constructive heuristic algorithm.

the preceding job, while the subsequent job or the planning horizon limits the latest start time. Then, the periods with lower electricity prices are searched preferentially within the time window to optimize the scheduling of the job blocks. The adjustment direction is primarily rightward, aiming to migrate job blocks from high price periods to periods with lower electricity prices. When the start time of a job block is adjusted, it may trigger passive shifts in subsequent job blocks. However, the adjustment process is strictly constrained to ensure that it does not exceed the time window of the subsequent job block and does not cause machine occupancy conflicts. If these conditions are violated, the current adjustment plan is abandoned. Due to fluctuations in electricity costs across different periods, adjusting the start time of each job block can reduce local electricity costs, thereby improving the overall quality of the solution. The specific process is shown in Algorithm 2.

Line 3 of Algorithm 2 identifies and records all job blocks on machine i . Line 5 applies a greedy strategy to identify feasible start time intervals for each job block and determines the start time that minimizes local energy consumption. Line 6 updates the start and end times of job block j . Finally, line 10 calculates the objective value of the current schedule.

4.2. Adaptive multi-population co-evolutionary algorithm

4.2.1. Framework of the proposed algorithm

Heuristic methods based solely on experience or simple rules often struggle to achieve best solutions. An adaptive multi-population co-evolutionary algorithm (AMCEA) is proposed to address this limitation. The algorithm solves three key sub-problems (machine selection,

Algorithm 2 Neighborhood search algorithm**Input:** Non-dominated solution set NS .**Output:** Non-dominated solution set NS' .

```

1: for  $g$  in  $G$  do
2:   for  $i$  in  $M_g$  do
3:     Record all job blocks on machine  $i$ ;  $\triangleright$  In time order from
       earliest to latest
4:     for each job block  $j$  on machine  $i$  do
5:       Find a feasible start time for job block  $j$  that minimizes
       energy cost;
6:       Update the start and end times of job block  $j$ ;
7:     end for
8:   end for
9: end for
10: Calculate objective values and update  $NS'$ .

```

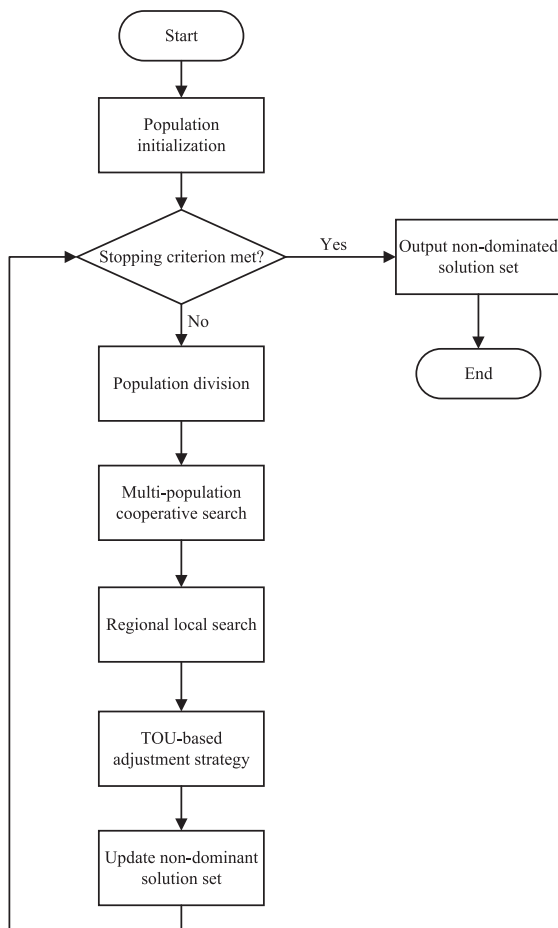


Fig. 3. The flowchart of the AMCEA algorithm.

job sequencing, and job grouping) through a two-stage coding strategy combined with heuristic rules. Uniformly distributed reference vectors are then used to divide the populations and promote information exchange among subpopulations through an adaptive cooperative strategy, thereby enhancing the diversity and coordination of the populations. In addition, two local reinforcement search operators are designed to adjust the search adaptively and enhance the convergence of the population. Finally, based on deterioration and non-deterioration completion times, two TOU tariff-based adjustment strategies are developed to reduce the total electricity cost. The framework of AMCEA is illustrated in Fig. 3.

Table 3

The energy consumption rates of machines.

	ep_i^1 or ep_i^2	ed_i^1 or ed_i^2	eo_i^1 or eo_i^2
MS_1	3	1	0
MS_2	4	1	0
MS_3	5	1	0
MS_4	4	1	0
MS_1^b	5	2	0
MS_2^b	6	2	0

Table 4

The information of jobs.

	p_j^1	p_j^2	s_j
J_1	2	6	3
J_2	3	5	2
J_3	1	4	4
J_4	1	3	4
J_5	3	3	5
J_6	2	4	2
J_7	3	6	3
J_8	2	4	4
J_9	3	5	4
J_{10}	2	3	4

	Stage 1										Stage 2									
JSV	3	4	1	6	8	10	2	7	9	5	3	4	1	10	6	8	7	2	5	9
MSV	1	3	2	3	2	4	4	1	2	3	1	1	1	2	1	2	2	1	2	1

Fig. 4. An encoded solution.

4.2.2. Encoding and decoding

The HFS-TOU problem requires solving two coupled subproblems: determining the job processing sequence and selecting machines for each job. To address this, a two-stage encoding is adopted (Zhang et al., 2008), consisting of a job sequence vector (JSV) and a machine selection vector (MSV), as illustrated in Fig. 4. Specifically, the solution is encoded into two parts, representing the job sequence and machine assignment for stage 1 and stage 2, respectively. In the JSV, elements correspond to job numbers, determining the processing sequence on each machine. In the MSV, elements represent the selected machines for each job, with a length equal to twice the total number of jobs.

The two-stage encoding addresses job sequencing and machine selection but does not directly handle grouping batches on batch processing machines. To address this, a method based on the FCFS rule and the FB rule is proposed. First, the FCFS rule is applied to sort jobs in non-descending order according to their release times in stage 1, giving higher priority to jobs with earlier release times. Then, the FB rule is applied to group jobs with similar completion times in stage 1 into the same batch without exceeding the capacity of the batch processing machine. This method aims to improve the utilization efficiency of batch processing machines.

An encoding example is shown in Fig. 4, with 10 jobs ($J_1, J_2, \dots, J_{10}$) and four machines (MS_1, MS_2, MS_3, MS_4) in stage 1. Stage 2 has two batch processing machines (MS_1^b, MS_2^b) with a capacity of seven. Machine and job details are in Tables 3 and 4, while the TOU electricity prices are shown in Fig. 5. The Gantt chart in Fig. 6 corresponds to the solution depicted in Fig. 4, with $C_{max} = 17$ and $TEC = 170.4$.

4.2.3. Population initialization

In population-based optimization, population initialization is crucial for achieving good optimization performance. For the HFS-TOU problem, initialization involves two key aspects: job sequencing and machine selection. Eight heuristic algorithms are developed to generate the initial population based on the encoding scheme. A high-quality initial population helps accelerate convergence and enhance search

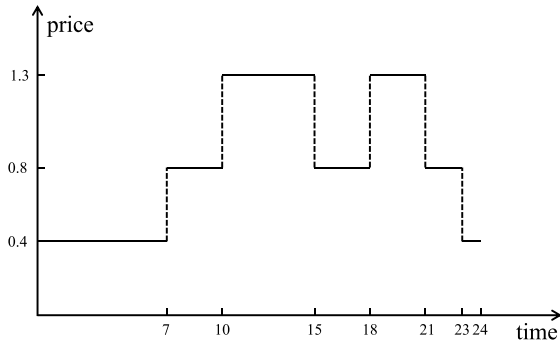


Fig. 5. TOU electricity pricing in a day.

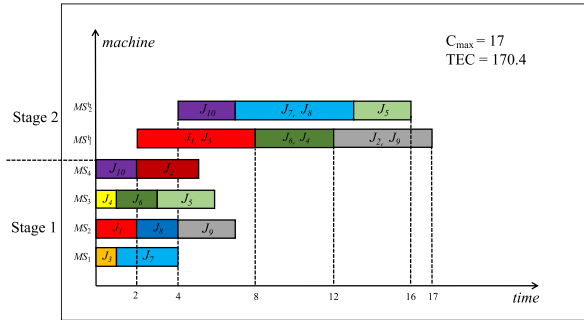


Fig. 6. The Gantt chart of a schedule.

effectiveness. Since job sequencing and machine assignment are coupled in HFS-TOU, combinations of sequencing and assignment rules are employed during initialization to enhance diversity and solution quality.

To generate the JSV, the following five rules are applied to determine the job processing sequence:

- (1) Johnson rule: This rule prioritizes jobs based on their processing times in the two stages, aiming to minimize the makespan by optimizing the job sequence.
- (2) Shortest processing time (SPT) rule: This rule prioritizes the job with the shortest processing time in stage 1.
- (3) Longest processing time (LPT) rule: This rule prioritizes the job with the longest processing time in stage 1.
- (4) Shortest total processing time (STP) rule: This rule prioritizes the job with the shortest total processing time in both stages, aiming to improve the overall production process efficiency.
- (5) Random rule: This rule generates a random sequence of jobs, aiming to increase diversity in the initial population.

To generate the MSV, the following three rules are applied to select machines for each job from its eligible machine set:

- (1) Minimum energy consumption (MEC) rule: This rule prioritizes the machine with the lowest energy consumption rate, aiming to reduce total energy costs.
- (2) Minimum used machine (MUM) rule: This rule prioritizes the machine with the fewest currently assigned tasks, aiming to balance the workload among the machines.
- (3) Random rule: This rule randomly selects a machine from the set of eligible machines, aiming to increase the diversity of machine selection.

With five JSV rules and three MSV rules, there are fifteen possible combinations of these heuristic rules. However, deterministic heuristics can lead to the aggregation of repeated solutions when subpopulations

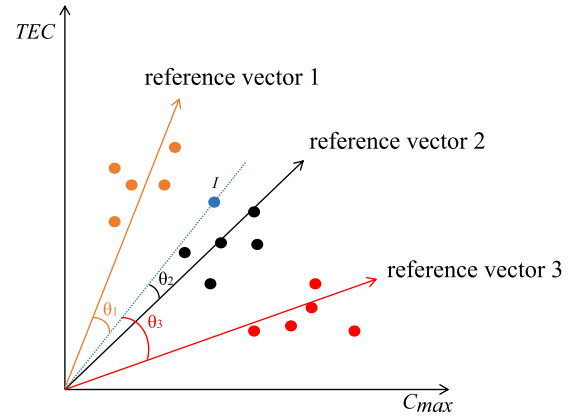


Fig. 7. Illustration of population division.

are divided, significantly reducing population diversity, and the diversity tends to deteriorate further after subpopulation partitioning. To address this, controlled random perturbations are introduced during the initial solution generation while preserving the heuristic guidance. Specifically, two types of perturbations are applied: (1) with a probability of 20%, two non-critical jobs in the sequence generated by the heuristic rule are randomly selected and swapped; and (2) during machine assignment, with a probability of 20%, a sub-best machine (e.g., the one with the second shortest processing time) is randomly selected instead of the machine recommended by the heuristic. Through this design, the generation process remains heuristic-based, while controllable randomness is introduced to break solution duplication. This strategy effectively maintains sufficient population diversity despite following deterministic heuristic rules.

4.2.4. Population division

To minimize errors caused by dimensional inconsistencies, it is necessary to eliminate the influence of different variable scales in the objective function. Normalizing the objective values ensures that all terms are dimensionally consistent, leading to more accurate optimization results. The specific normalization process is illustrated in Eq. (40). As this study addresses a bi-objective problem, the x -th solution in generation t is denoted as $F_x^t = \{F_{x,1}^t, F_{x,2}^t\}$, where $F_{x,o}^t$ represents the value of individual x on objective o . The maximum and minimum objective values in generation t are denoted as $Z_{max}^t = \{Z_{max,1}^t, Z_{max,2}^t\}$ and $Z_{min}^t = \{Z_{min,1}^t, Z_{min,2}^t\}$, respectively. The normalized objective value is $F_{x,o}^t$. The set of individuals in generation t is denoted as P^t .

$$F_{x,o}^t = \frac{F_{x,o}^t - Z_{min,o}^t}{Z_{max,o}^t - Z_{min,o}^t} \quad x \in P^t, o \in \{1, 2\} \quad (40)$$

For each x , the cosine of the angle between its normalized objective value vector and each reference vector v is calculated by Eq. (41). Then, individual x is assigned to the subpopulation corresponding to the reference vector v with the minimum cosine angle value between its objective value vector. Here, $|V|$ denotes the number of uniformly distributed vectors in the reference vector set V (Das & Dennis, 1998). In Fig. 7, the yellow, black, and red vectors represent three reference vectors, and the cosines of the angles θ_1 , θ_2 , and θ_3 between individual x and the three reference vectors are calculated, respectively. Individual x is assigned to the subpopulation corresponding to the reference vector with the minimum cosine, θ_2 . The other individuals are assigned similarly, and the entire population is divided into subpopulations 1, 2, and 3, corresponding to the yellow, black, and red subpopulations in the figure, respectively.

$$\cos \theta_{x,v}^t = \frac{F_x^t \cdot v}{\|F_x^t\| \cdot \|v\|}, \quad x \in P^t, v \in V. \quad (41)$$

Based on $|V|$ reference vectors, the population P^t is divided into $|V|$ subpopulations. The resulting distribution of individuals reflects the solution distribution in the search space and guides the cooperative selection of pairing partners.

4.2.5. Adaptive cooperative search

In multi-objective optimization problems, the pairing selection strategy of individuals significantly influences algorithm performance. Traditional methods often determine pairing partners through simple random selection. However, the lack of targeting in random pairing makes it difficult for the offspring distribution to evolve toward the desired direction. The random pairing impedes the efficient combination and transmission of superior genes, thereby limiting both the diversity and convergence of individuals. To address this issue, this paper designs an adaptive cooperative search strategy based on the distribution characteristics of the solution set. The strategy incorporates two pairing modes: intra-subpopulation selection and inter-subpopulation selection. Individuals within the same subpopulation exhibit higher solution similarity, strengthening local search capability. In contrast, individuals across different subpopulations possess greater diversity, thereby promoting global search ability. Subpopulations are divided according to distinct reference vectors. This adaptive strategy dynamically balances exploration and exploitation, thereby improving the overall search performance of the algorithm.

An adaptive probability adjustment mechanism is proposed to dynamically adjust the selection probabilities of the two pairing methods to adapt to the algorithm's search requirements at different stages. In the early stage of the cooperative search, greater emphasis is placed on global exploration to promote broader traversal of the solution space. In contrast, local exploitation becomes gradually more important in the middle and later stages. Based on this idea, the cooperative selection probability is dynamically adjusted during evolution, assigning different probabilities in different stages to meet the search demands better. This mechanism helps balance solution development and exploration throughout the evolutionary process. The detailed procedure is shown in Algorithm 3.

Algorithm 3 Adaptive adjustment probability

Input: Current iteration number t ; Total number of iterations t_{max} .

Output: Cooperative selection probability pr_t .

```

1:  $sum \leftarrow 0$ .
2: for  $k = 1 : t - 1$  do
3:    $value(k) \leftarrow \frac{n_1(k)}{n_1(k) + n_2(k)}$ .
4:    $sum \leftarrow sum + value(k)$ .
5: end for
6: if  $sum \leq 0.2 \cdot t_{max}$  then
7:    $pr_t \leftarrow p_{r_{min}}$ .
8: else if  $sum \geq 0.8 \cdot t_{max}$  then
9:    $pr_t \leftarrow p_{r_{max}}$ .
10: else
11:    $pr_t \leftarrow p_{r_{max}} - \frac{(p_{r_{max}} - p_{r_{min}}) \cdot sum}{t - 1}$ .
12: end if
```

In Algorithm 3, lines 3 to 4 calculate the cumulative contribution of new solutions up to generation $t - 1$. Here, n_1 represents the number of intra-subpopulation pairings, and n_2 represents the number of inter-subpopulation pairings. Lines 6 to 11 then adjust the probabilities for intra- and inter-subpopulation pairing for the next iteration. The thresholds $0.2t_{max}$ and $0.8t_{max}$ were determined based on preliminary analysis and empirical consideration to achieve a balance between exploration and exploitation. Specifically, $0.2t_{max}$ in line 6 corresponds to the early stage of the algorithm, where the probability is set to $p_{r_{min}}$ to encourage broader exploration through inter-subpopulation pairing. In contrast, $0.8t_{max}$ in line 8 indicates the later stage, where the probability is set to $p_{r_{max}}$ to promote intensive local exploitation via intra-subpopulation

pairing. Throughout evolution, the probabilities are dynamically tuned based on the cumulative contribution of different pairing types, ensuring a balanced transition between exploration and exploitation. The updated probabilities guide the cooperative search in the next iteration.

Algorithm 4 Adaptive cooperative search

Input: Current population P^t ; Population size N_p ; Current iteration number t ; Total number of iterations t_{max} .

Output: New population P^t .

```

1: Computer cooperative selection probability  $pr_t$ .
2: for  $i = 1 : N_p$  do
3:   if  $\text{rand}() \leq pr_t$  then
4:     Select individual  $x_{new}$  from the same subpopulation as individual  $x_i$ .
5:   else
6:     Select individual  $x_{new}$  from the different subpopulation than individual  $x_i$ .
7:   end if
8:    $r \leftarrow \text{Crossover}(x_i, x_{new})$ .
9:    $r_{new} \leftarrow \text{Mutation}(r)$ .
10: end for
```

In Algorithm 4, for each individual x_i in the current population, if a randomly generated probability is less than or equal to the cooperative selection probability pr_t , a new individual x_{new} is selected from the same subpopulation as x_i ; otherwise, x_{new} is selected from a different subpopulation. After the pairing partners are determined, adaptive crossover and mutation are applied to generate new candidate solutions.

Prior to crossover, the fitness values $f(x_p)$ of both parent solutions are calculated using Eq. (42), where the Chebyshev aggregation function (Qi et al., 2014) is defined in Eq. (43). Here, λ_p is the weight vector of the subpopulation to which x_p belongs, z^* represents the reference point of the entire population (with each element being the minimum value across all solutions for the corresponding objective function), and $F(x_p)$ denotes the objective function values of x_p . Let f' denote the higher fitness value of the two parent solutions. The crossover probability p_c is adaptively determined based on f' , the maximum population fitness f_{max} , and the average population fitness f_{avg} , as described by Eq. (44). The settings $p_{c_{min}} = 0.5$ and $p_{c_{max}} = 0.9$ are empirically determined based on preliminary experiments.

$$f(x_p) = \frac{1}{g(x_p | \lambda_p, z^*)} \quad (42)$$

$$g(x_p | \lambda_p, z^*) = \max_{l=1,2} \{ \lambda_p^l \cdot |F_l(x_p) - z_l^*| \} \quad (43)$$

$$p_c = \begin{cases} p_{c_{min}} + \frac{(p_{c_{max}} - p_{c_{min}}) \cdot (f' - f_{avg})}{f_{max} - f_{avg}}, & f' \geq f_{avg} \\ p_{c_{max}}, & f' < f_{avg} \end{cases} \quad (44)$$

After determining the crossover probability p_c , the crossover operation is performed via the Position-Based Crossover (PBX) operator, as shown in Fig. 8. Specifically, a binary string of the same length as the JSV is first randomly generated for the two parent chromosomes. When the value at a given position in the binary string is 1, the corresponding gene from the parent is reserved for one offspring; when it is 0, the corresponding gene from the other parent is assigned. Moreover, it must also be ensured that the MSV gene corresponding to the retained JSV gene is preserved simultaneously, and the remaining genes are inherited sequentially from the parent that did not retain the gene. This process is repeated to produce two offspring, after which one offspring is randomly selected for subsequent mutation operations.

The mutation operator adopts a position-based mutation approach. The mutation probability p_m is first calculated using Eq. (45), where $p_{m_{min}} = 0.1$ and $p_{m_{max}} = 0.3$ are empirically set. If the generated probability satisfies p_m , the mutation operation is independently performed on each stage as follows:

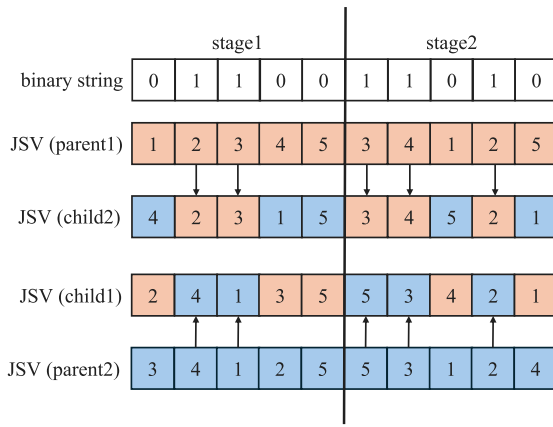


Fig. 8. An example of Position-based Crossover (PBX).

- (1) Select one job randomly from each stage.
- (2) Change the assigned machine of the selected job to another eligible machine.

$$p_m = \begin{cases} p_{m_{\min}} + \frac{(p_{m_{\max}} - p_{m_{\min}})(f' - f_{\text{avg}})}{f_{\max} - f_{\text{avg}}}, & f' \geq f_{\text{avg}} \\ p_{m_{\max}}, & f' < f_{\text{avg}} \end{cases} \quad (45)$$

4.2.6. Regional local search

This study focuses on a bi-objective optimization problem and proposes two local search operators based on the two objectives:

The first operator (denoted by LS1) aims to reduce the total makespan. It randomly selects a job or batch on the critical path and assigns it to another qualified machine with the least workload. If the new assignment reduces the makespan, it is accepted; otherwise, it remains unchanged.

The second operator (denoted by LS2) aims to reduce total energy consumption. It randomly selects a job from the machine with the highest load and transfers it to a machine with the lowest load. This targeted selection strategy is based on the observation that machines with higher load typically accumulate greater energy consumption, thus offering more potential for improvement. By focusing on jobs assigned to the most heavily loaded machines, LS2 increases the likelihood of identifying relocations that meaningfully reduce total energy costs. In contrast, random job selection across all machines often involves jobs already on low-load machines, leading to ineffective moves and wasted search effort. Furthermore, redistributing jobs from high-load to low-load machines contributes to load balancing, which can shorten the completion times and minimize energy consumption. If the new assignment reduces energy consumption, it is accepted; otherwise, it remains unchanged.

LS1 and LS2 are designed to focus on different but interrelated optimization objectives. Specifically, LS1 aims to reduce the makespan, while LS2 primarily targets minimizing total energy consumption. However, since energy consumption is strongly influenced by the makespan, through factors such as machine operating periods and idle energy, LS2 inherently considers both objectives during the local search process. As the population evolves, the relative importance of makespan and energy consumption changes dynamically. To adaptively utilize these two operators, a region-based selection strategy is proposed. As illustrated in Fig. 9, the objective space is uniformly divided into three regions using reference points. In Region I, LS2 is applied to reduce energy consumption. In Region II, LS1 is adopted to minimize the makespan.

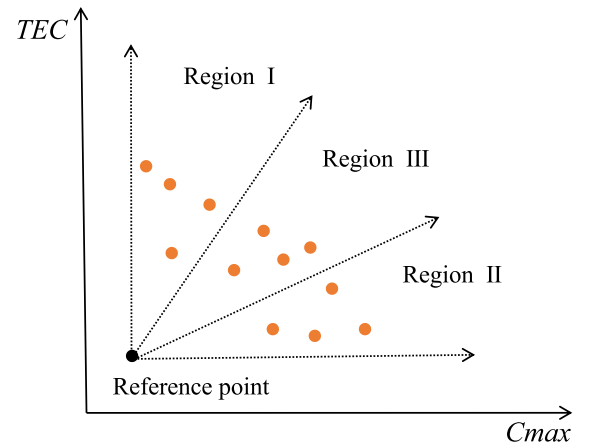


Fig. 9. The illustration of division in objective space.

To avoid local optima, LS1 or LS2 is randomly selected with equal probability in Region III. The detailed local search strategy is presented in Algorithm 5. In line 2, the parameter *Loop* controls the search depth, which is determined according to the settings in Section 5.3. Moreover, a random selection operator is used during the first generation of the evolutionary process.

Algorithm 5 Regional local search

Input: Current population P^t ; Size of population N_p ; Regional search depth *Loop*.

Output: New population P^{t+1} .

```

1: for  $i = 1 : N_p$  do
2:   for  $j = 1 : \text{Loop}$  do
3:     if Individual  $x_i$  is located in Region I then
4:       Apply LS2 to individual  $x_i$  to generate  $x_i^{\text{new}}$ .
5:     else if Individual  $x_i$  is located in Region II then
6:       Apply LS1 to individual  $x_i$  to generate  $x_i^{\text{new}}$ .
7:     else
8:       Randomly apply LS1 or LS2 to generate individual  $x_i^{\text{new}}$ .
9:     end if
10:    if Individual  $x_i^{\text{new}}$  dominates individual  $x_i$  then
11:       $x_i \leftarrow x_i^{\text{new}}$ .
12:    end if
13:    Update the non-dominated solution set  $N.S.$ 
14:  end for
15: end for

```

4.2.7. TOU-based adjustment

Two adjustment strategies based on time-of-use (TOU) electricity pricing are proposed:

- (1) Strategy 1: Reduce the power cost by adjusting the job to be processed from the high-price period to the low-price period without increasing the completion time. The search direction is from back to front to find the best processing start time, resulting in lower energy consumption.
- (2) Strategy 2: Reduce the total energy consumption by decreasing the completion time. However, excessively extending the completion time can negatively affect productivity and incur additional electricity costs during machine idleness. Given that the TOU electricity tariff function is periodic, the scheduling search is extended by one more period ($\Delta = 24$ h). The energy consumption is optimized by adjusting the job's start time to find the best processing start time over the extended time horizon.

Algorithm 6 TOU-based adjustment**Input:** Non-dominated solution set NS .**Output:** Non-dominated solution set NS' .

```

1: for  $x$  in  $NS$  do
2:   for  $g$  in  $G$  do
3:     for  $i$  in  $M_g$  do
4:       for  $j$  in jobs (batches, if  $g=2$ ) on machine  $i$  do ▷ From back to front
5:         if  $g = 1$  then
6:           Calculate the start time  $NT_j$  of each  $j$  in stage 2.
7:         else
8:            $NT_j \leftarrow C_{max}$ .
9:         end if
10:        Calculate the start time  $ST_j$ , processing time  $PT_j$  and completion time  $CT_j$  for each  $j$ .
11:         $val_{min} \leftarrow 0$ ,  $val \leftarrow 0$ .
12:        while  $CT_j + val < \min\{ST_{j+1}, NT_j\}$  do
13:          if  $EC(ST_j, val_{min}) > EC(ST_j, val)$  then
14:             $val_{min} \leftarrow val$ .
15:          end if
16:           $val \leftarrow val + 1$ .
17:        end while
18:        if  $val_{min} > 0$  then
19:           $ST_j \leftarrow ST_j + val_{min}$ .
20:           $CT_j \leftarrow CT_j + val_{min}$ .
21:        end if
22:      end for
23:    end for
24:    Form new individual  $x^{new}$ .
25:  end for
26:  if  $x^{new}$  dominates  $x$  then
27:     $x^{new} \leftarrow x$ .
28:  end if
29: end for

```

The framework for Strategy 1 is presented in Algorithm 6. The pseudo-code for Strategy 2 is similar but with an extended search range; thus, it is omitted here.

In Algorithm 6, the non-dominated solution set NS serves as both the input and output, with the TOU electricity price adjustment applied to the NS obtained from the previous iteration. Lines 1 to 10 traverse the jobs (or batches) on each machine in reverse order, computing the start time, processing time, and completion time for each job or batch. Lines 11 to 25 aim to locally minimize energy consumption by calculating the backward delayed processing interval val for each job (or batch). Finally, in lines 26 to 29, the non-dominated solution set is refined by updating inferior solutions with improved ones, thereby enhancing the overall quality of the solution set.

5. Computational experiments

This section evaluates the performance of the proposed algorithms through a series of experiments on test instances with different sizes. Section 5.1 introduces the experimental data generation. Section 5.2 presents the evaluation metrics used. Section 5.3 describes the parameter settings for the algorithms. Sections 5.4 and 5.5 report the results of ablation experiments and comparative experiments, respectively. The algorithms are implemented in MATLAB 2021a, and Gurobi 11 is used to solve the MIP model. All experiments are conducted on a computer with an Intel Core i7 CPU (2.90 GHz) and the 64-bit Windows 10 operating system.

5.1. Experiment data

Following the experimental setup in Jia et al. (2019) and Wang et al. (2020), random test instances are generated by combining different levels of key parameters. Although Wang et al. (2020) proposed classic

benchmark instances for HFS problems, these instances do not consider critical features addressed in this study, such as machine eligibility constraints, batch processing, or TOU electricity pricing. Similarly, Jia et al. (2019) incorporated energy optimization considerations, but their problem assumes variable machine speeds, which differ from the fixed-speed batch processing machines considered here. Therefore, based on the experimental setups of these studies, the instance generation process is adapted to accommodate the specific constraints and characteristics of the proposed problem. The specific parameter settings used for generating test instances are listed in Table 5.

- n : Number of jobs.
- s_j : Size of each job.
- p_j^1, p_j^2 : Processing times in stage 1 and 2.
- R : Machine eligibility proportion in stage 1.
- Q : Capacity of batch processing machines in stage 2.
- m_1, m_2 : Number of machines in stages 1 and 2.
- ep_j^1, ep_j^2 : Machine processing energy rates in stages 1 and 2.
- ed_j^1, ed_j^2 : Machine idle energy rates in stages 1 and 2.
- $f(t)$: Function of TOU electricity price, by the guidelines from Ding et al. (2015), is depicted in Table A.1, and it is further simplified as Eq. (46).

To comprehensively cover different problem combinations, 25 sets of test instance parameters were generated using a mixed orthogonal design, as shown in Table 6. In the parameter setting experiment described in Section 5.3, one instance was generated for each set of parameters. Five instances were generated for each set of parameters in the ablation experiments (Section 5.4) and the comparative experiments (Section 5.5).

Table 5
Experimental parameter settings.

Factors	Values	Number of levels
n	{10, 20, 50, 100, 200}	5
s_j	$U\{1, Q\}$	1
p_j^1	$U\{1, 3\}$	1
p_j^2	$U\{3, 6\}$	1
R	{0.6, 0.8, 1}	3
Q	{7, 10}	2
m_1	{3, 5, 10}	3
m_2	{2, 5}	2
ep_j^1	$U\{3, 5\}$	1
eo_j^1	$U\{1, 2\}$	1
ep_j^2	$U\{4, 6\}$	1
eo_j^2	$U\{2, 3\}$	1

Table 6
Relevant parameters of the 25 sets of test instances.

No.	n	m_1	m_2	Q	R
I_1	10	3	2	7	1
I_2	10	5	2	10	0.8
I_3	10	5	5	7	0.8
I_4	10	10	5	10	0.6
I_5	10	3	2	7	0.6
I_6	20	3	2	7	1
I_7	20	5	5	7	0.8
I_8	20	3	5	10	0.8
I_9	20	5	2	7	0.6
I_{10}	20	10	2	10	0.6
I_{11}	50	10	5	7	1
I_{12}	50	3	5	7	0.8
I_{13}	50	5	2	10	0.6
I_{14}	50	5	2	7	0.6
I_{15}	50	3	2	10	0.8
I_{16}	100	3	2	10	0.8
I_{17}	100	3	2	7	0.6
I_{18}	100	10	2	7	0.8
I_{19}	100	5	5	7	0.6
I_{20}	100	5	5	10	1
I_{21}	200	3	5	7	0.6
I_{22}	200	3	5	10	0.6
I_{23}	200	5	2	10	1
I_{24}	200	5	2	7	0.8
I_{25}	200	10	2	7	0.8

$$f(h) = \begin{cases} 0.4, & \text{if } 0 \leq h < 7 \\ 0.8, & \text{if } 7 \leq h < 10 \\ 1.3, & \text{if } 10 \leq h < 15 \\ 0.8, & \text{if } 15 \leq h < 18 \\ 1.3, & \text{if } 18 \leq h < 21 \\ 0.8, & \text{if } 21 \leq h < 23 \\ 0.4, & \text{if } 23 \leq h < 24 \end{cases} \quad (46)$$

5.2. Evaluation metrics

To evaluate the performance of multi-objective optimization algorithms, three evaluation metrics are used to measure the quality of the obtained non-dominated solution sets:

- (1) Number of non-dominated solutions (NNS, Qian et al. (2020)). NNS measures the diversity of the non-dominated solution set found by an algorithm. Generally, more non-dominated solutions indicate better algorithm effectiveness.
- (2) Hypervolume (HV, Zitzler et al. (2003)). HV evaluates how well the non-dominated solution set covers the objective space, indicating the algorithm's overall performance. A larger HV implies

Table 7
Parameter levels.

Level	N_p	$ V $	$p_{r_{max}}$	$p_{r_{min}}$	Loop
1	100	10	0.9	0.1	1
2	200	20	0.8	0.2	2
3	300	30	0.7	0.3	3
4	400	40	0.6	0.4	4
5	500	50	0.5	0.5	5

the algorithm has better convergence and diversity and is closer to the true Pareto front.

- (3) Coverage (C, Zitzler and Thiele (1999)). C compares two algorithms by calculating the proportion of solutions in one set dominated by the other. $C(X, Y)$ is the proportion of solutions in Y dominated by X. A higher $C(X, Y)$ means X covers more of the Pareto front than Y.
- (4) Runtime. Running time as an evaluation metric is usually used to compare the time performance of different algorithms under the same population size, the same number of iterations, and other identical conditions.

5.3. Parameter settings

The proposed AMCEA algorithm has five key parameters: population size N_p , number of reference vectors $|V|$, upper and lower limits of pairing probability $p_{r_{max}}$ and $p_{r_{min}}$, and the regional search depth *Loop*.

To determine the optimal parameter configuration of the AMCEA algorithm, Taguchi's method (Roy, 2001) was used for parameter optimization. As shown in Table 7, the values of each parameter have five levels. Based on the design principles of Taguchi's experiment, 25 different sets of parameter combinations were designed, and the specific experimental settings are shown in Table A.2.

To evaluate and select the optimal parameter combination, we used the average value of the HV metrics of the non-dominated solution set obtained from the test instances in Table 6 as the experimental response value. The experimental results were processed using the Minitab 21 statistical analysis tool and presented using the mean effect plot in Fig. 10. The horizontal axis in the figure represents the different parameter settings, and the vertical axis corresponds to the mean value of HV. Higher values imply better performance for the corresponding parameter settings. Analyzing Fig. 10 leads to the optimal combination of parameter configurations for the AMCEA algorithm as follows: $N_p=300$, $|V|=20$, $p_{r_{max}}=0.8$, $p_{r_{min}}=0.4$ and *Loop*=3.

5.4. Ablation experiments

Ablation experiments (Ren et al., 2016) are conducted to validate the effectiveness of strategies used in the proposed AMCEA algorithm. Three modified versions of the AMCEA algorithm are designed:

- AMCEA-NL: Without regional local search.
- AMCEA-NT: Without TOU-based adjustment.
- AMCEA-NR: LS2 in regional local search is replaced with a strategy that randomly selects jobs.

All three improved algorithms were executed 10 times on 25 test instances, which were generated according to the parameter settings of the I_{21} to I_{25} instance groups, with five instances created for each group. For example, I_{21-1} denotes the first instance in the I_{21} group. The average values of the evaluation metrics across the 10 runs were then calculated for comparison purposes. The ablation experiments were revised by introducing a unified termination condition to ensure a fair and consistent basis for performance comparison. Specifically, the runtime of AMCEA-NL, which does not employ local search, was adopted as the common termination criterion for all algorithms, thereby enabling comparison under the same time limit. Furthermore,

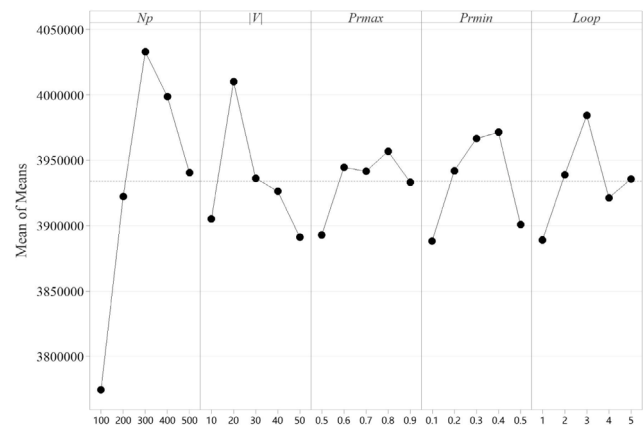


Fig. 10. Results of main effects of Means.

Table 8
Results of ablation experiments in NNS and HV on the test instances with 200 jobs, where “+”, “−” and “~” indicate that an algorithm is better, worse, and similar to AMCEA, respectively, and “+/-/~” indicates the number of instances where an algorithm is better/worse/similar to AMCEA.

Instance	AMCEA		AMCEA-NL		AMCEA-NT		AMCEA-NR	
	NNS	HV	NNS	HV	NNS	HV	NNS	HV
I ₂₁₋₁	3.1	469586.93	8.1	46710.00−	7.9	48342.69−	3.8	5782.30−
I ₂₁₋₂	2.4	502401.60	8.6	55172.40−	7.4	54718.17−	4.2	9859.01−
I ₂₁₋₃	2.0	593678.80	8.2	99518.85−	6.5	95671.83−	5.7	23864.89−
I ₂₁₋₄	2.0	531342.30	10.8	72152.69−	9.4	68508.53−	5.7	16734.19−
I ₂₁₋₅	7.2	81578.86	6.4	76116.08−	5.1	15487.52−	3.3	20361.80−
I ₂₂₋₁	6.3	60966.62	5.0	25075.74−	4.8	42911.19−	4.5	13152.90−
I ₂₂₋₂	6.6	62204.84	5.7	17312.77−	5.5	48539.78−	4.1	7510.60−
I ₂₂₋₃	7.0	61604.26	6.0	39071.08−	7.4	43248.52−	4.8	25419.61−
I ₂₂₋₄	7.5	69515.83	5.2	9963.05−	7.1	66045.63−	4.4	14575.03−
I ₂₂₋₅	5.6	68219.35	6.8	45283.35−	5.3	25199.11−	5.1	7399.79−
I ₂₃₋₁	5.7	121703.76	8.1	109162.44−	6.2	96078.51−	4.8	98059.89−
I ₂₃₋₂	5.7	103180.37	10.7	80319.22−	5.2	69462.87−	4.6	87104.52−
I ₂₃₋₃	6.5	86313.11	11.1	74473.90−	10.4	70892.33−	5.1	76964.27−
I ₂₃₋₄	6.3	150016.30	7.2	138603.44−	7.1	141196.14−	3.5	136157.40−
I ₂₃₋₅	7.4	98980.53	8.8	89522.32−	6.7	88416.08−	3.2	90251.15−
I ₂₄₋₁	5.5	84019.08	3.7	53346.68−	4.7	57511.83−	6.4	24582.32−
I ₂₄₋₂	4.0	80945.30	1.0	49764.00−	5.2	37919.35−	3.5	24045.73−
I ₂₄₋₃	4.0	79300.80	1.0	48105.20−	4.1	34805.16−	5.7	15857.15−
I ₂₄₋₄	7.2	62894.60	1.0	48105.20−	4.6	36766.43−	6.7	19516.21−
I ₂₄₋₅	7.1	96989.64	1.0	82940.00−	4.0	60987.35−	4.7	43743.74−
I ₂₅₋₁	5.1	41734.90	4.1	28374.16−	11.1	37026.09−	4.4	35955.75−
I ₂₅₋₂	7.7	53954.53	5.5	45029.47−	7.0	50474.71−	5.4	47376.33−
I ₂₅₋₃	9.1	99171.85	3.4	84479.29−	9.3	90454.03−	3.5	74255.43−
I ₂₅₋₄	6.1	80595.29	4.7	50798.62−	13.4	62977.44−	2.9	63788.76−
I ₂₅₋₅	5.0	81272.54	4.8	52186.23−	12.9	65981.12−	3.1	63553.47−
+/-/~			0/25/0		0/25/0		0/25/0	

the Wilcoxon signed-rank test with a significance level of 0.05 was conducted on the results of the 10 runs to assess whether the performance differences between algorithms are statistically significant. The ablation results are reported in Tables 8 and 9, where the best metric values for each instance are highlighted in bold.

In Table 8, the test instances with 200 jobs are selected for analysis, and the results of AMCEA, AMCEA-NL, AMCEA-NT, and AMCEA-NR on NNS and HV are compared to evaluate the impact of different strategies on AMCEA’s performance. The results show that: (1) with respect to HV, AMCEA performs significantly better than AMCEA-NL, AMCEA-NT, and AMCEA-NR on all 25 instances, indicating that the regional local search strategy, TOU-based adjustment strategy, and LS2 strategy substantially enhance the convergence of the proposed algorithm, thereby verifying both the effectiveness and necessity of these strategies. It should be noted that the only difference between AMCEA and AMCEA-NR is whether a deterministic job selection strategy is introduced in LS2. AMCEA-NR randomly selects jobs from any machine and assigns them to the machine with the lowest load. In contrast, AMCEA employs a load-oriented heuristic strategy that selects

jobs only from the machine with the highest load, aiming to mitigate machine imbalance and enhance the optimization of TEC. Although this modification is structurally simple, experimental results demonstrate that it is effective in reducing TEC. Theoretically, the random selection strategy may also lead to high-quality solutions. However, due to its lack of guidance, the optimization results rely more on chance and tend to be less stable. In this study, the TEC exhibits a wide value range and high sensitivity to small perturbations in the solution space, making purely random local search unlikely to consistently locate solutions with low TEC. In contrast, although the deterministic heuristic strategy may reduce the diversity of the solution set to some extent, it can more effectively guide the search towards regions with lower energy consumption, thereby improving the overall solution quality in terms of TEC. (2) regarding NNS, although AMCEA did not achieve the largest number of non-dominated solutions on all instances, it is able to generate a relatively large number of solutions on most instances. This result suggests that the three strategies contribute to maintaining the diversity of AMCEA. In addition, this phenomenon may be attributed to the small size of the true Pareto front on certain instances, which

Table 9
Results of ablation experiments in C on the test instances with 200 jobs.

Instance	C(AMCEA, AMCEA-NL)	C(AMCEA-NL, AMCEA)	C(AMCEA, AMCEA-NT)	C(AMCEA-NT, AMCEA)	C(AMCEA, AMCEA-NR)	C(AMCEA-NR, AMCEA)
I_{21-1}	1	0	1	0	1	0
I_{21-2}	1	0	1	0	1	0
I_{21-3}	1	0	1	0	1	0
I_{21-4}	1	0	1	0	1	0
I_{21-5}	0.59	0.24	0.50	0	0.11	0
I_{22-1}	0.41	0.07	0.49	0.15	0.11	0
I_{22-2}	0.46	0.03	0.61	0.08	0.18	0
I_{22-3}	0.33	0.11	0.55	0.17	0.04	0
I_{22-4}	0.37	0	0.55	0.26	0.29	0
I_{22-5}	0.80	0	0.37	0.11	0.22	0
I_{23-1}	0.52	0.08	0.74	0	0.53	0
I_{23-2}	0.77	0.17	0.52	0.11	0.28	0.13
I_{23-3}	0.23	0.05	0.51	0.03	0.47	0.17
I_{23-4}	0.31	0.03	0.14	0.13	0.39	0.06
I_{23-5}	0.37	0	0.23	0.09	0.44	0.08
I_{24-1}	1	0	0.70	0.1	0.61	0.05
I_{24-2}	1	0	0.79	0	0.40	0
I_{24-3}	1	0	0.71	0	0.59	0
I_{24-4}	1	0	0.76	0	0.69	0
I_{24-5}	1	0	0.57	0	0.45	0
I_{25-1}	0.48	0.13	0.61	0.21	0.62	0.38
I_{25-2}	0.77	0.21	0.67	0.15	0.80	0.01
I_{25-3}	0.92	0.01	0.49	0.27	0.87	0.09
I_{25-4}	0.45	0.06	0.51	0.37	0.27	0.09
I_{25-5}	0.50	0.11	0.49	0.32	0.44	0.05

leads to an increased number of identical solutions as the algorithm converges, ultimately reducing the number of non-dominated solutions obtained. In summary, the comparative results fully demonstrate that the regional local search strategy, the TOU-based adjustment strategy, and the LS2 strategy effectively enhance the problem-solving capability of AMCEA.

Furthermore, the comparison results of AMCEA and its three variants in terms of the C metric are presented in Table 9. As shown in the results, AMCEA outperforms the other three algorithms on all test instances, demonstrating that the regional local search strategy, TOU-based adjustment strategy, and LS2 strategy significantly improve the convergence of the proposed algorithm. In addition, comparison of AMCEA with each variant in terms of the C metric reveals that, compared with AMCEA-NR, AMCEA shows a more pronounced advantage over AMCEA-NL and AMCEA-NT. This indicates that the regional local search and TOU-based adjustment strategies are more critical in enhancing algorithmic performance, thereby verifying their effectiveness and importance.

5.5. Comparative experiments

To validate the proposed CH and AMCEA algorithms, the ϵ -constraint method (Bérubé et al., 2009) in Gurobi 11 is utilized to solve the MIP model and obtain the exact Pareto front for small instances. The ϵ -constraint method transforms constraints into the objective function, converting the problem into a single-objective optimization with an augmented objective. To limit computation time, Gurobi's runtime is restricted to 3600 s. Since this study is based on the paper by Wang et al. (2020) and extended in a tempered glass production environment, the Tabu Search (TS) and Ant Colony Optimization (ACO) algorithms from that paper are used for comparison. In addition, the IMOEA/D algorithm (Wu & Cao, 2022), which is related to solving the hybrid flow shop problem containing batch processing machines, is also used for comparison experiments. The hyperparameter settings for all compared algorithms follow the corresponding papers. The test instances are generated based on the parameter settings of the I_1 to I_{25} instance groups, with five instances created for each group, resulting on 125 instances. The results of the evaluation metrics are averaged over 10 independent runs, and the best result for each instance is highlighted in bold. In addition, the Wilcoxon signed-rank test with a significance

level of 0.05 is conducted on the results of the 10 runs to statistically verify the significance of performance differences between algorithms.

In Table 10, the results of Gurobi, AMCEA, CH, IMOEA/D, TS and ACO algorithms are listed on small-scale test instances with 10 and 20 jobs in terms of NNS and HV indicators, where Gurobi corresponds to the results obtained within 3600 s. Based on the given results, it can be seen that (1) overall, AMCEA can obtain the best performance on most test instances, among which its HV values on 38 test instances are significantly better than Gurobi; (2) CH and IMOEA/D perform second overall, where the former has strong competitiveness on several small-scale instances, and the latter has relatively stable performance in NNS. At the same time, the overall performance of TS and ACO is not as good as that of other methods. They did not achieve the highest HV value on all instances, and the diversity and quality of their solutions were obviously insufficient. (3) AMCEA not only has obvious advantages in HV but also performs well in NNS, which is maintained at a high level on average on each instance, further indicating that the algorithm can maintain good diversity while obtaining high-quality solution sets, taking into account both accuracy and breadth. This result verifies that AMCEA can effectively solve small-scale optimization problems.

Since Gurobi cannot obtain valid solutions within 3600 s when the number of jobs exceeds 20, only the results of AMCEA and its counterparts on NNS and HV metrics are given in Tables 11 and 12. According to Table 11, (1) regarding the HV metric, AMCEA delivered the best performance on most instances. Specifically, it achieved the highest HV values in 84 out of 100 test instances, demonstrating clear superiority in convergence and diversity. Notably, IMOEA/D obtained the best HV results in I_{20-3} and I_{20-5} . Moreover, although IMOEA/D yielded higher HV values on some instances, statistical tests revealed no significant differences from AMCEA. (2) Regarding the NNS metric, CH and IMOEA/D generated a larger number of non-dominated solutions on most instances, reflecting better diversity of solutions. Although AMCEA did not outperform them in this regard, its superior HV performance suggests that it places greater emphasis on enhancing solution quality and convergence while maintaining acceptable diversity. In contrast, CH and IMOEA/D, though capable of producing more solutions, are relatively weaker in terms of convergence. In addition, TS only achieved the best HV result on instance I_{12-1} , with its overall performance falling short of AMCEA. The ACO algorithm performed the worst across all test instances, showing no competitive advantage.

Table 10

Comparative results in NNS and HV on the test instances with 10 and 20 jobs, where “/” represents that Gurobi cannot obtain any feasible solution in 3600 s.

Instance	Gurobi			AMCEA		CH		IMOE/D		TS		ACO	
	NNS	HV	Gap(%)	NNS	HV	NNS	HV	NNS	HV	NNS	HV	NNS	HV
I ₁₋₁	11	11906.25	17.83	6.6	14176.49	1.0	6737.50–	6.5	13893.31–	7.7	10189.93–	3.1	7286.60–
I ₁₋₂	5	15636.55	0	5.0	15262.36	5.0	13478.45–	5.6	15375.46~	2.0	12995.95–	3.6	6363.20–
I ₁₋₃	12	23136.90	20.93	5.5	28289.57	1.0	23014.65–	6.0	26294.25~	6.6	20973.33–	4.0	10212.55–
I ₁₋₄	6	25875.15	14.11	7.3	27588.56	6.0	22473.75–	4.8	26360.72–	1.0	23466.75–	5.7	11265.75–
I ₁₋₅	5	32537.93	14.46	7.2	34304.59	5.0	27944.83–	4.8	33590.56–	4.6	29177.23–	5.1	21718.88–
I ₂₋₁	3	24477.90	23.26	3.4	26429.20	1.0	23664.50–	4.4	25840.35–	1.0	20792.80–	6.6	17086.20–
I ₂₋₂	6	20448.80	23.94	2.4	22747.01	3.0	19274.10–	3.4	21935.10–	1.0	18202.05–	7.5	13207.65–
I ₂₋₃	3	30464.95	0	3.0	20681.80	5.0	19564.30–	3.7	20465.40~	3.7	17471.77–	3.3	10863.20–
I ₂₋₄	8	21863.28	26.37	2.9	26004.04	3.0	20487.08–	3.6	25409.13–	4.9	20269.97–	1.0	8552.78–
I ₂₋₅	9	27138.25	13.32	6.3	28510.22	3.0	24629.70–	6.8	28081.99–	6.8	20744.98–	1.0	10918.85–
I ₃₋₁	7	16068.35	22.71	7.3	18700.53	3.0	15826.80–	6.3	18451.10–	4.2	14408.82–	2.5	8953.05–
I ₃₋₂	7	18850.50	22.63	5.7	20681.80	5.0	19564.30–	7.7	20465.40~	3.4	17471.77–	1.0	10863.20–
I ₃₋₃	4	21360.63	7.58	5.2	22595.65	5.0	20247.48–	6.2	22426.11–	8.4	17867.15–	3.2	8621.28–
I ₃₋₄	7	19465.50	19.68	6.3	22350.65	3.0	19598.80–	5.9	22196.18–	3.7	19646.36–	1.0	9851.40–
I ₃₋₅	9	21088.20	38.71	7.1	23684.95	3.0	21274.25–	6.2	23698.11~	7.3	18892.73–	5.5	10078.55–
I ₄₋₁	1	48707.10	17.35	1.5	48211.26	2.0	47324.10–	1.3	47583.88–	1.0	43492.75–	3.2	27011.50–
I ₄₋₂	1	39067.60	21.54	1.2	38443.64	1.0	37506.00–	1.6	37992.96–	1.0	35832.35–	1.0	5692.20–
I ₄₋₃	1	33582.68	18.25	1.9	33379.67	2.0	33261.68–	1.5	32789.31–	3.0	30940.23–	1.0	5126.43–
I ₄₋₄	1	46575.00	23.49	1.9	46441.76	1.0	44500.70–	2.3	45758.84–	1.0	42564.60–	3.0	10110.80–
I ₄₋₅	1	36102.93	14.13	1.3	35414.58	1.0	34603.88–	1.1	35172.39–	1.0	32995.63–	2.7	6837.08–
I ₅₋₁	3	22793.45	13.32	2.8	23474.46	2.0	20412.50–	1.1	25020.65+	1.0	18174.45–	5.4	8183.70–
I ₅₋₂	3	14026.05	14.84	2.0	14389.37	2.0	12962.35–	2.0	14458.05~	1.0	10631.75–	2.3	5421.90–
I ₅₋₃	2	31700.35	15.04	3.2	31497.41	2.0	28241.35–	2.7	31105.65~	1.0	28604.70–	5.1	9595.00–
I ₅₋₄	2	25022.90	5.58	3.0	24385.27	2.0	21565.95–	2.0	24391.54~	1.0	21315.15–	5.6	8806.90–
I ₅₋₅	5	17827.20	21.54	3.2	18323.90	2.0	15729.15–	3.1	17736.43–	2.0	15812.75–	3.3	6764.60–
I ₆₋₁	6	40618.50	15.04	2.0	46305.90	2.0	32975.60–	2.3	46939.30~	1.2	33511.33–	2.9	11923.10–
I ₆₋₂	5	31600.85	15.94	2.6	42109.95	3.0	24191.90–	2.2	42532.46~	1.0	22400.55–	2.8	14029.00–
I ₆₋₃	6	77741.85	7.83	2.1	95976.08	1.0	62515.40–	1.0	95408.87~	1.1	55570.58–	9.1	35121.30–
I ₆₋₄	3	44331.80	18.25	2.1	56583.98	3.0	41112.50–	1.1	56065.52–	1.0	39698.08–	2.0	12810.20–
I ₆₋₅	/	/	/	2.1	61405.36	4.0	47162.90–	1.7	61053.66~	1.5	45362.90–	6.2	16955.25–
I ₇₋₁	2	99139.60	50.36	5.3	100497.64	4.0	91752.30–	4.8	100083.82~	1.0	89729.40–	4.8	25514.30–
I ₇₋₂	2	90162.78	47.38	5.9	91203.34	4.0	83082.93–	5.0	90906.32–	1.0	80993.25–	6.4	26911.53–
I ₇₋₃	2	125501.35	50.36	7.2	128232.35	4.0	117113.25–	4.9	127569.68–	1.0	114199.80–	4.4	35463.00–
I ₇₋₄	2	110907.48	50.36	7.4	112262.32	4.0	103085.33–	5.1	112127.02~	1.0	100777.50–	7.1	33224.48–
I ₇₋₅	2	97915.45	50.50	7.1	100038.92	4.0	90498.90–	4.6	99749.01~	1.0	87861.83–	4.3	23391.85–
I ₈₋₁	2	66461.08	66.18	11.9	87411.68	9.0	84733.83–	6.4	76077.78–	1.4	71094.47–	2.0	26738.18–
I ₈₋₂	2	79784.13	47.39	5.9	82894.95	10.0	94092.88+	11.8	92360.23+	1.9	73603.12–	3.1	24275.18–
I ₈₋₃	3	103182.08	51.02	11.8	109921.37	11.0	107757.58–	6.1	98150.07–	2.2	89946.09–	5.2	34293.48–
I ₈₋₄	/	/	/	12.2	108459.13	9.0	106891.08–	7.6	94656.45–	2.7	86982.22–	4.7	31710.38–
I ₈₋₅	/	/	/	11.2	83390.84	8.0	78375.23–	7.5	72857.10–	2.7	65386.45–	1.6	23651.68–
I ₉₋₁	3	68900.33	29.10	4.7	77851.24	1.0	59004.93–	4.2	66003.02–	1.0	60572.62–	8.2	26454.43–
I ₉₋₂	3	48950.10	28.50	5.0	56485.54	5.0	43555.45–	4.7	45703.27–	3.7	44314.58–	4.1	15952.70–
I ₉₋₃	4	51688.75	20.34	3.9	58697.06	1.0	43876.35–	4.3	48241.26–	1.2	42781.83–	3.7	15168.80–
I ₉₋₄	4	76070.90	23.43	4.9	87114.08	1.0	60240.75–	3.9	75462.65–	1.7	65490.37–	4.1	28151.20–
I ₉₋₅	2	62089.80	25.95	4.6	74722.71	1.0	53851.20–	4.5	63274.10–	2.1	55850.20–	2.1	23329.55–
I ₁₀₋₁	5	61111.43	33.25	5.3	66721.61	2.0	53230.38–	3.9	58328.82–	3.6	54255.49–	5.3	20192.68–
I ₁₀₋₂	4	83609.78	31.97	5.7	92488.04	1.0	72985.13–	4.8	81286.66–	2.0	73137.03–	5.4	36361.08–
I ₁₀₋₃	6	75817.35	32.29	4.8	84698.62	1.0	65213.70–	4.2	75041.90–	1.2	63400.13–	6.7	29180.25–
I ₁₀₋₄	5	67968.23	32.02	4.8	73854.77	1.0	57864.38–	4.5	64718.76–	2.4	57533.36–	5.5	24166.68–
I ₁₀₋₅	8	45577.65	30.32	5.0	49492.27	1.0	33901.40–	4.8	42412.63–	3.8	41065.20–	3.2	15631.80–
+ / - / ~						1/49/0		2/35/13		0/50/0		0/50/0	

Furthermore, the results on large-scale test instances with 200 jobs are presented in Table 12. It can be observed that: (1) AMCEA still shows a clear advantage in terms of the HV metric. Among the 25 instances, AMCEA achieved the highest HV values in 21 instances and ranked second in the remaining four, confirming its superiority across problems of different scales. In I_{23-2} , one of the two instances where IMOE/D obtained the best HV results, AMCEA showed no significant difference, demonstrating its effectiveness and competitiveness. (2) With respect to the NNS metric, AMCEA achieved the highest value on four instances. In comparison, CH and IMOE/D performed best on 10 instances each, indicating they provided more substantial solution diversity on certain instances. TS and ACO did not achieve the best HV results on any instance, and their overall performance was relatively poor. Similar to the observations in Table 11, although AMCEA did not dominate in NNS, it demonstrated a clear advantage in solution quality and convergence while maintaining a reasonable level of diversity.

Furthermore, the results of AMCEA, CH, IMOE/D, TS, and ACO in terms of the C metric on the test instances with 10, 20, 50, 100,

and 200 jobs are presented in Tables 13–15, respectively. The results show that: (1) across all test instances, AMCEA is significantly better than TS and ACO, and outperforms these two algorithms on nearly all instances; (2) AMCEA also demonstrates significant superiority over CH and IMOE/D on most instances, with its advantage becoming even more pronounced as the number of jobs increases. These results confirm that AMCEA effectively solves scheduling optimization problems at various scales, particularly for large-scale instances.

As shown in Fig. 11, the CH algorithm achieves faster runtime than the other comparison algorithms on most test instances. This is mainly attributed to its rapid computation speed, interpretability, and the extensive use of heuristics, which enable it to perform well in solving complex problems. Overall, AMCEA does not dominate in terms of runtime. Its performance is slightly better than that of IMOE/D and ACO, but slower than CH and TS.

In summary, for the hybrid flow shop energy-efficient scheduling problem considering TOU electricity prices and batch processors with identical capacities, the proposed AMCEA algorithm outperforms

Table 11

Comparative results in terms of NNS and HV on the test instances with 50 and 100 jobs.

Instance	AMCEA		CH		IMOE/D		TS		ACO	
	NNS	HV	NNS	HV	NNS	HV	NNS	HV	NNS	HV
I ₁₁₋₁	2.6	130999.33	3.0	114020.15~	3.3	124678.52~	3.3	116826.56~	5.1	56670.65~
I ₁₁₋₂	3.2	129214.39	8.0	115508.60~	3.7	124227.88~	7.5	102624.29~	2.3	45526.70~
I ₁₁₋₃	3.6	293652.49	19.0	290004.28~	3.5	282875.80~	13.6	248388.74~	4.7	134906.68~
I ₁₁₋₄	3.3	198960.85	19.0	199805.53~	3.0	191726.84~	15.2	162449.97~	3.4	89164.48~
I ₁₁₋₅	2.3	296381.74	21.0	270739.78~	1.9	290792.00~	19.5	242201.20~	6.2	166809.68~
I ₁₂₋₁	4.4	101500.38	8.0	102897.12~	4.6	95229.44~	8.9	104959.26~	3.3	63282.08~
I ₁₂₋₂	5.4	87068.64	10.0	100261.90+	6.4	83085.45~	8.4	86450.30~	1.4	31508.10~
I ₁₂₋₃	4.5	135989.88	8.0	134723.10~	5.5	130801.42~	8.7	130044.49~	2.2	49387.90~
I ₁₂₋₄	5.3	115831.90	8.0	123864.48+	6.0	110632.47~	7.2	107214.04~	2.5	39890.83~
I ₁₂₋₅	6.1	102229.47	11.0	102912.60~	5.7	97052.26~	4.7	88974.23~	2.9	39682.80~
I ₁₃₋₁	5.9	155079.96	2.0	110525.00~	7.1	144388.06~	2.0	122041.42~	5.1	74611.35~
I ₁₃₋₂	3.0	46684.55	3.0	47978.60+	4.8	46026.29~	1.2	43558.59~	1.8	17175.90~
I ₁₃₋₃	2.7	40840.54	3.0	42541.69+	4.6	39942.95~	1.3	37545.04~	1.6	16368.39~
I ₁₃₋₄	3.3	91563.26	2.0	86795.70~	4.6	89154.09~	2.5	76979.54~	1.0	34599.60~
I ₁₃₋₅	4.0	98306.26	10.0	85476.32~	5.1	95952.51~	1.3	91861.79~	2.0	35026.17~
I ₁₄₋₁	4.2	211312.67	4.0	182551.78~	6.3	197350.81~	5.9	161730.86~	3.9	102559.43~
I ₁₄₋₂	5.0	226162.78	5.0	217305.65~	6.4	209517.52~	2.3	179448.27~	5.2	110442.80~
I ₁₄₋₃	5.1	137323.88	8.0	105640.75~	7.9	125818.36~	1.4	101238.40~	2.3	45842.53~
I ₁₄₋₄	4.5	192269.45	6.0	185749.70~	6.3	178253.34~	4.1	146664.16~	3.0	87033.10~
I ₁₄₋₅	3.3	106259.83	5.0	100627.34~	7.6	98700.39~	1.7	78629.89~	1.0	44727.76~
I ₁₅₋₁	2.4	129672.79	3.0	116512.88~	7.6	98471.20~	2.7	107802.38~	4.1	51683.28~
I ₁₅₋₂	5.9	144497.78	11.0	118364.23~	6.0	136415.01~	2.8	111609.83~	1.0	47615.88~
I ₁₅₋₃	3.1	113911.33	5.0	110925.08~	5.3	102385.47~	2.4	86978.30~	2.2	58015.50~
I ₁₅₋₄	3.7	86281.45	4.0	78686.05~	6.1	68000.78~	4.4	68149.94~	1.0	39686.10~
I ₁₅₋₅	5.2	98725.71	8.0	99261.78~	7.6	74255.68~	2.2	83136.00~	3.5	42095.63~
I ₁₆₋₁	3.8	786329.53	7.0	442335.23~	3.6	729788.61~	5.0	394119.98~	7.1	205119.98~
I ₁₆₋₂	3.2	852307.05	4.0	444636.95~	1.8	820855.77~	4.0	424885.40~	6.3	187572.55~
I ₁₆₋₃	3.0	818932.58	4.0	417194.83~	7.1	405003.71~	4.0	398138.83~	5.6	199682.18~
I ₁₆₋₄	2.1	622263.45	4.0	351773.43~	7.4	308560.52~	1.0	195836.38~	3.0	135896.53~
I ₁₆₋₅	3.2	442245.62	7.0	342014.25~	8.2	306026.84~	1.0	228680.85~	4.8	147097.25~
I ₁₇₋₁	3.2	548588.29	4.0	486249.40~	8.4	497491.80~	2.0	366771.85~	5.4	251596.30~
I ₁₇₋₂	3.9	665710.81	7.0	603466.63~	10.4	688353.47~	3.0	374234.18~	7.5	273572.73~
I ₁₇₋₃	6.4	520466.24	10.0	537209.80~	9.2	515994.24~	1.0	282717.90~	3.3	233693.40~
I ₁₇₋₄	7.8	411456.58	4.0	415691.78~	9.5	356344.01~	3.0	230565.88~	3.0	201273.08~
I ₁₇₋₅	2.8	387434.52	6.0	411824.13+	10.5	369961.36~	2.0	230549.53~	4.6	178679.13~
I ₁₈₋₁	5.4	1589512.60	10.0	1565335.90~	10.2	1562792.66~	3.0	1192355.90~	8.2	758303.15~
I ₁₈₋₂	5.8	909709.37	8.0	683691.90~	8.5	877074.58~	5.0	537367.70~	2.0	437584.90~
I ₁₈₋₃	4.4	1285455.16	16.0	1127085.33~	9.1	1226084.13~	9.0	783712.83~	4.3	573986.98~
I ₁₈₋₄	4.4	1334106.70	7.0	1141749.55~	9.2	1268364.07~	3.0	763721.35~	7.4	564921.75~
I ₁₈₋₅	4.2	1067400.63	3.0	921183.20~	7.3	986285.22~	2.0	629700.75~	6.1	445894.75~
I ₁₉₋₁	5.0	541360.32	10.0	526778.15~	8.9	512271.12~	6.0	472832.95~	4.7	227252.20~
I ₁₉₋₂	4.9	565904.32	7.0	551238.48~	9.7	519422.80~	6.0	485921.18~	3.0	232864.98~
I ₁₉₋₃	3.3	734464.31	15.0	716959.50~	8.2	676585.80~	6.0	620644.40~	5.1	264760.85~
I ₁₉₋₄	3.9	568329.07	12.0	545502.00~	7.3	521500.08~	9.0	459120.75~	3.2	240827.15~
I ₁₉₋₅	3.7	644792.76	12.0	642146.20~	7.9	581916.43~	6.0	533037.90~	4.7	209195.30~
I ₂₀₋₁	6.8	709306.63	18.0	612687.60~	6.8	708455.12~	6.0	551649.35~	3.0	196856.00~
I ₂₀₋₂	4.7	792576.14	18.0	674396.50~	6.1	79837.21~	7.0	658762.80~	3.3	202981.40~
I ₂₀₋₃	6.7	756081.51	16.0	694944.58~	5.9	764835.94+	5.0	623523.33~	4.2	183857.88~
I ₂₀₋₄	5.3	729582.48	17.0	596732.95~	6.1	746826.49~	6.0	620189.65~	3.9	195605.40~
I ₂₀₋₅	4.6	759769.62	16.0	613765.65~	5.8	777423.88+	12.0	631757.70~	3.2	205937.65~
+/-/~			5/38/7		2/40/8		0/48/2		0/50/0	

IMOE/D, TS, and ACO in terms of solution quality in most cases. In contrast, the CH algorithm performs better in terms of running time.

5.6. Impact of TOU and flat rate pricing on scheduling decisions

The experiments above verify the effectiveness of the proposed model and algorithm. This section further explores the impact of electricity pricing mechanisms on peak load reduction through a comparative analysis of different pricing mechanisms in scheduling schemes, thereby providing managerial insights for actual production organizations and energy management.

The experiment is based on instance I_{16-1} to more accurately explore the influence of different electricity prices and pricing structures on the scheduling scheme, where the limitation on processing machines for jobs in the first stage is removed, allowing each job to be processed on all machines. At the same time, the processing energy consumption rate of each machine in both the first and second stages is set to the same value to eliminate the influence of factors other than electricity

pricing. Two electricity pricing strategies are designed to evaluate the guiding effect of different electricity pricing mechanisms on electricity cost control and peak load reduction while keeping the property values of jobs and machines consistent. The first is the TOU pricing of industrial electricity in Beijing. The second is a flat rate strategy, which sets three electricity prices based on the TOU: the maximum value of TOU (TOU_{max} , FR1), the average value (TOU_{avg} , FR2), and the minimum value (TOU_{min} , FR3) within 24 h.

Table 16 presents the results under the four electricity pricing strategies, including the value range of the two objectives, the HV values, and the peak load reduction rates in both stages. The peak load reduction rate is the ratio of the total number of jobs during off-peak electricity periods (summed over each unit hour) to the total number of jobs across all periods. Considering that FR2 represents the average electricity price of the TOU strategy, Fig. 12 compares the Pareto frontiers obtained under TOU and FR2. Furthermore, to analyze distribution differences in peak load, Fig. 13 illustrates the number of jobs over time in the two stages for two scheduling solutions with

Table 12

Comparative results in terms of NNS and HV on the test instances with 200 jobs.

Instance	AMCEA		CH		IMOEA/D		TS		ACO	
	NNS	HV	NNS	HV	NNS	HV	NNS	HV	NNS	HV
I ₂₁₋₁	3.1	2621653.56	10.0	1392875.55–	5.2	1392303.35–	5.0	971856.60–	4.3	671247.35–
I ₂₁₋₂	2.4	2924222.10	8.0	1479800.85–	4.7	1514345.28–	3.0	1256823.20–	5.7	614420.45–
I ₂₁₋₃	2.0	2735376.30	6.0	1411550.25–	4.7	1415039.24–	1.0	1101922.50–	2.0	574789.60–
I ₂₁₋₄	2.0	2765908.05	6.0	1439043.90–	8.2	1394940.33–	4.0	1047468.60–	3.5	667644.35–
I ₂₁₋₅	7.2	1306454.80	13.0	1308106.38~	7.4	1259372.30–	2.0	915643.88–	3.0	563205.73–
I ₂₂₋₁	6.3	4003091.00	5.0	3769628.40–	6.6	3916309.50–	5.0	3726679.70–	6.3	1046321.90–
I ₂₂₋₂	6.6	3662137.74	7.0	3550798.60–	5.4	3192932.84–	5.0	3408372.20–	5.1	1018827.90–
I ₂₂₋₃	7.0	2115075.53	6.0	2029376.77–	4.8	1823275.34–	8.0	1601856.39–	3.2	657734.57–
I ₂₂₋₄	7.5	1097725.23	11.0	1095946.56~	6.0	866486.02–	5.0	663095.66–	4.2	533505.91–
I ₂₂₋₅	5.6	1091619.43	11.0	1057977.90–	5.6	823120.94–	4.0	753887.73–	2.8	563891.78–
I ₂₃₋₁	5.7	3166677.02	3.0	3174033.30+	5.7	2962816.53~	1.0	2256747.60–	3.1	786634.60–
I ₂₃₋₂	5.7	3566767.85	5.0	3238716.43~	5.9	3663107.86~	1.0	2095815.38–	7.2	972554.98–
I ₂₃₋₃	6.5	3530717.40	6.0	3487426.05–	4.0	3624410.10+	3.0	2256925.25–	5.5	955506.00–
I ₂₃₋₄	6.3	3764445.19	6.0	3702167.80–	3.3	3634733.16–	3.0	2858605.25–	4.2	1036430.05–
I ₂₃₋₅	7.4	3721705.75	10.0	3575198.60–	5.4	3699419.38~	1.0	2955308.80–	6.7	948861.00–
I ₂₄₋₁	5.5	3636597.93	10.0	3211631.70–	6.5	3610157.45~	5.0	1712507.90–	3.1	903714.50–
I ₂₄₋₂	4.0	3546992.05	7.0	2943022.15–	8.8	3193065.98–	1.0	2014331.55–	3.3	923636.95–
I ₂₄₋₃	4.0	3765829.53	4.0	3272495.68–	9.0	3397926.04–	4.0	1838815.93–	4.7	1039745.78–
I ₂₄₋₄	7.2	3690845.80	4.0	3131009.55–	9.6	3356717.62~	2.0	1751322.35–	3.2	945235.95–
I ₂₄₋₅	7.1	3690563.69	12.0	3208453.10–	6.6	3366946.29–	1.0	2079165.90–	4.6	1004067.45–
I ₂₅₋₁	5.1	4451427.83	7.0	3496905.80–	7.8	4207638.70–	2.0	2658387.80–	3.4	1128133.90–
I ₂₅₋₂	7.7	5407252.34	3.0	4849083.95–	5.4	5074288.39–	3.0	3095243.75–	3.8	1320693.35–
I ₂₅₋₃	9.1	4707206.28	4.0	3530674.18–	11.9	4377580.13–	4.0	2807559.98–	2.0	1208999.03–
I ₂₅₋₄	6.1	4585793.39	6.0	4462109.20–	10.4	4305341.28–	3.0	2757854.10–	3.0	1319262.00–
I ₂₅₋₅	5.0	4264174.44	9.0	4053618.7–	7.2	4026446.91–	5.0	2388703.10–	3.5	1141934.10–
+/-/~				1/22/2	1/23/1		0/25/0		0/25/0	

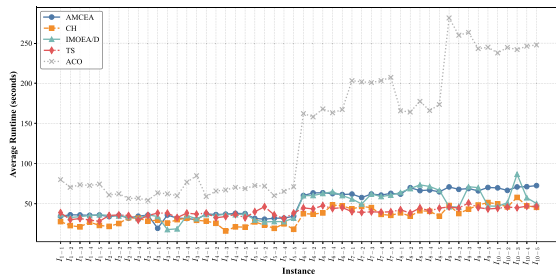
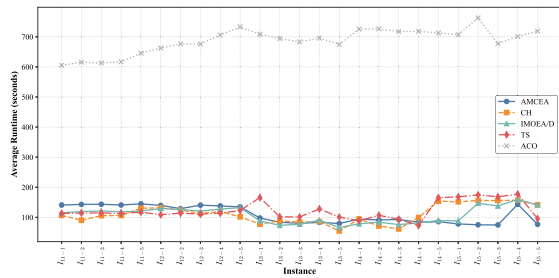
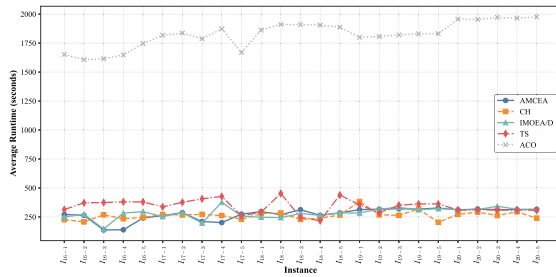
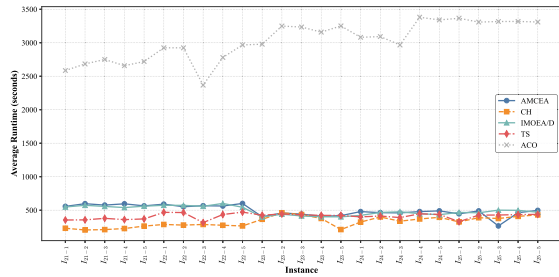
(a) Comparison of runtime of different algorithms on instance groups I_1-I_{10} .(b) Comparison of runtime of different algorithms on instance groups $I_{11}-I_{15}$.(c) Comparison of runtime of different algorithms on instance groups $I_{16}-I_{20}$.(d) Comparison of runtime of different algorithms on instance groups $I_{21}-I_{25}$.

Fig. 11. Comparison of runtime of different algorithms on instance groups.

similar makespan ($C_{\max}$), highlighted by red circles in Fig. 12. In Fig. 13, the line graph represents the electricity price curve, while the bar chart depicts the number of jobs in each time interval.

The comparison between TOU and FR2 shows that the dynamic electricity pricing strategy is more effective in regulating peak loads and promoting energy-efficient scheduling. Under the TOU pricing scheme, the peak load reduction rates in the first and second stages reached 68.83% and 67.95%, respectively, higher than the corresponding values of 67.86% and 66.28% under the FR2 pricing scheme. In addition, the HV value of the scheduling solution obtained under TOU

is slightly higher than that under FR2, indicating better convergence and diversity. Furthermore, the TEC range under TOU is narrower than that of FR2. Referring to Fig. 13, it can be observed that TOU enables more effective utilization of off-peak periods.

By comparing the scheduling schemes under the three fixed-rate pricing strategies (FR1, FR2, and FR3), the sensitivity of scheduling performance to the design of flat rates can be further evaluated. As shown in Table 16, although FR1 achieved relatively high peak load reduction rates (68.12% in the first stage and 66.92% in the second stage), its HV value dropped significantly, suggesting poor convergence

Table 13
Comparative results in terms of C on the test instances with 10 and 20 jobs.

Instance	C(AMCEA, CH)	C(CH, AMCEA)	C(AMCEA, IMOEA/D)	C(IMOEA/D, AMCEA)	C(AMCEA, TS)	C(TS, AMCEA)	C(AMCEA, ACO)	C(ACO, AMCEA)
I ₁₋₁	1	0	0.65	0.26	1	0	1	0
I ₁₋₂	1	0	0.38	0.41	1	0	1	0
I ₁₋₃	0.90	0.03	0.67	0.52	1	0	1	0
I ₁₋₄	1	0	0.28	0.54	0.90	0	1	0
I ₁₋₅	1	0	0.21	0.49	1	0	1	0
I ₂₋₁	1	0	0.93	0.03	1	0	1	0
I ₂₋₂	1	0	0.83	0.13	1	0	1	0
I ₂₋₃	0.60	0.03	0.36	0.32	1	0	0.94	0
I ₂₋₄	1	0	1	0	1	0	1	0
I ₂₋₅	1	0	0.52	0.29	1	0	1	0
I ₃₋₁	1	0	0.48	0.34	1	0	1	0
I ₃₋₂	1	0	0.70	0.15	1	0	1	0
I ₃₋₃	1	0	0.67	0.26	1	0	1	0
I ₃₋₄	1	0	0.52	0.30	1	0	1	0
I ₃₋₅	1	0	0.36	0.57	1	0	1	0
I ₄₋₁	0.50	0	0.90	0	1	0	1	0
I ₄₋₂	1	0	0.90	0.10	1	0	1	0
I ₄₋₃	0.35	0.25	0.85	0.05	1	0	1	0
I ₄₋₄	1	0	0.85	0.10	1	0	1	0
I ₄₋₅	1	0	0.90	0.10	1	0	1	0
I ₅₋₁	1	0	0.20	0.80	1	0	1	0
I ₅₋₂	1	0	0.15	0.60	1	0	1	0
I ₅₋₃	1	0	0.2833	0.56	1	0	1	0
I ₅₋₄	1	0	0.4	0.49	1	0	1	0
I ₅₋₅	1	0	0.6	0.32	1	0	1	0
I ₆₋₁	1	0	0	0.55	1	0	1	0
I ₆₋₂	1	0	0.2333	0.34	1	0	1	0
I ₆₋₃	1	0	0	0.07	1	0	1	0
I ₆₋₄	1	0	0.05	0.07	1	0	1	0
I ₆₋₅	1	0	0.32	0.07	1	0	1	0
I ₇₋₁	0.80	0.11	0.79	0.13	0.50	0.09	1	0
I ₇₋₂	0.83	0.07	0.84	0.10	0.50	0.04	1	0
I ₇₋₃	0.70	0.14	0.78	0.13	0.70	0.03	1	0
I ₇₋₄	0.75	0.07	0.69	0.10	0.50	0.04	1	0
I ₇₋₅	0.80	0.10	0.77	0.13	0.30	0.04	1	0
I ₈₋₁	0.97	0.02	0.83	0.01	1	0	1	0
I ₈₋₂	0	0.78	0.03	0.81	1	0.014	1	0
I ₈₋₃	0.38	0.51	0.78	0.03	1	0	1	0
I ₈₋₄	0.81	0.15	0.86	0.02	1	0	1	0
I ₈₋₅	0.70	0.25	0.78	0.03	1	0	1	0
I ₉₋₁	1	0	0.80	0	1	0	1	0
I ₉₋₂	1	0	0.94	0	1	0	1	0
I ₉₋₃	1	0	0.88	0	1	0	1	0
I ₉₋₄	1	0	0.88	0.02	1	0	1	0
I ₉₋₅	1	0	0.93	0	1	0	1	0
I ₁₀₋₁	1	0	0.79	0.13	1	0	1	0
I ₁₀₋₂	1	0	0.71	0.11	1	0	1	0
I ₁₀₋₃	1	0	0.77	0.12	1	0	1	0
I ₁₀₋₄	1	0	0.83	0.15	1	0	1	0
I ₁₀₋₅	1	0	0.98	0.03	1	0	1	0

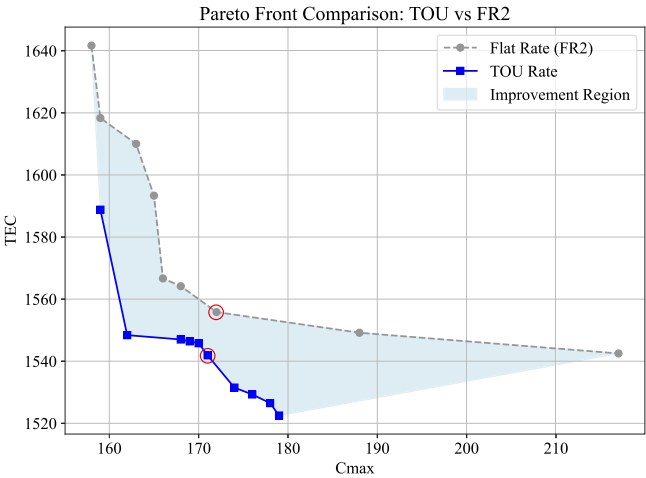


Fig. 12. Comparison of Pareto fronts under TOU and FR2.

Table 14
Comparative results in terms of C on the test instances with 50 and 100 jobs.

Instance	C(AMCEA, CH)	C(CH, AMCEA)	C(AMCEA, IMOEA/D)	C(IMOEA/D, AMCEA)	C(AMCEA, TS)	C(TS, AMCEA)	C(AMCEA, ACO)	C(ACO, AMCEA)
I ₁₁₋₁	1	0	1	0	1	0	1	0
I ₁₁₋₂	0.99	0	1	0	1	0	1	0
I ₁₁₋₃	0.59	0	1	0	1	0	1	0
I ₁₁₋₄	0.47	0	1	0	1	0	1	0
I ₁₁₋₅	1	0	1	0	1	0	1	0
I ₁₂₋₁	0.53	0	0.89	0.025	0.30	0.19	0.73	0
I ₁₂₋₂	0.10	0.25	0.95	0	0.65	0	1	0
I ₁₂₋₃	0.63	0	0.98	0	0.91	0.03	1	0
I ₁₂₋₄	0.54	0.10	0.97	0	0.97	0.02	1	0
I ₁₂₋₅	0.27	0	1	0	0.93	0	1	0
I ₁₃₋₁	1	0	0.95	0.01	1	0	1	0
I ₁₃₋₂	0.47	0.08	0.39	0.03	0.8	0.08	1	0
I ₁₃₋₃	0.43	0.18	0.27	0.10	0.6	0.11	1	0
I ₁₃₋₄	0.90	0	0.79	0	1	0	1	0
I ₁₃₋₅	1	0	0.60	0	0.5	0.19	1	0
I ₁₄₋₁	1	0	0.72	0.04	0.9	0	1	0
I ₁₄₋₂	0.60	0	0.93	0.02	1	0	1	0
I ₁₄₋₃	1	0	0.84	0.03	1	0	1	0
I ₁₄₋₄	0.83	0	1	0	1	0	1	0
I ₁₄₋₅	0.66	0.02	0.69	0	1	0	1	0
I ₁₅₋₁	1	0	1	0	1	0	1	0
I ₁₅₋₂	0.80	0	0.19	0.4752	0.7	0.04	1	0
I ₁₅₋₃	0.72	0.04	0.90	0.05	1	0	1	0
I ₁₅₋₄	0.50	0	1	0	0.9	0	1	0
I ₁₅₋₅	0.75	0	1	0	1	0	1	0
I ₁₆₋₁	1	0	1	0	1	0	1	0
I ₁₆₋₂	1	0	0	0	1	0	1	0
I ₁₆₋₃	1	0	1	0	1	0	1	0
I ₁₆₋₄	1	0	1	0	1	0	1	0
I ₁₆₋₅	0.47	0	1	0	0.3	0	1	0
I ₁₇₋₁	1	0	1	0	1	0	1	0
I ₁₇₋₂	0.56	0	0.65	0	0.77	0	1	0
I ₁₇₋₃	0.60	0	0.45	0.14	1	0	1	0
I ₁₇₋₄	0.58	0	0.99	0	1	0	1	0
I ₁₇₋₅	0.55	0	0.71	0	0	0	1	0
I ₁₈₋₁	0.69	0	0.70	0	0.37	0	1	0
I ₁₈₋₂	1	0	0.59	0	1	0	1	0
I ₁₈₋₃	1	0	1	0	0.78	0	1	0
I ₁₈₋₄	1	0	0.86	0	1	0	1	0
I ₁₈₋₅	1	0	1	0	0.60	0	1	0
I ₁₉₋₁	0.74	0.02	0.70	0.03	0.73	0	1	0
I ₁₉₋₂	0.69	0	0.69	0	0.92	0	1	0
I ₁₉₋₃	0.60	0.34	0.94	0	0.92	0	1	0
I ₁₉₋₄	0.73	0	0.87	0	0.87	0	1	0
I ₁₉₋₅	0.53	0	0.73	0	1	0	1	0
I ₂₀₋₁	0.44	0.26	0.85	0	0.40	0	1	0
I ₂₀₋₂	0.93	0	0.83	0	0.14	0.23	1	0
I ₂₀₋₃	0.61	0.19	0.80	0	0.20	0.11	1	0
I ₂₀₋₄	0.77	0	0.81	0	0.17	0	1	0
I ₂₀₋₅	0.93	0	0.75	0.02	0.17	0	1	0

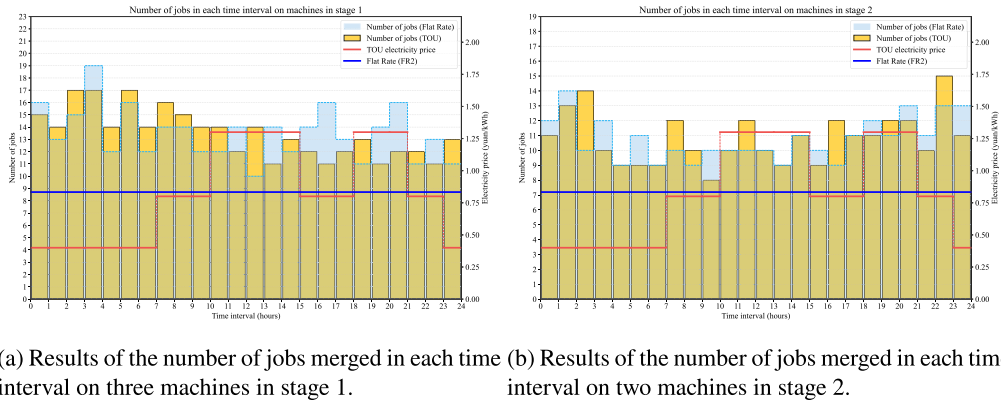


Fig. 13. Results of the number of jobs in each time interval during the two stages under FR2..

and diversity of the non-dominated solutions. In contrast, FR3 achieved the highest HV value, indicating better performance in terms of solution

quality. The HV value of FR2 lies between FR1 and FR3, but its peak load reduction rates are lower than those of FR1 and FR3. These results

Table 15
Comparative results in terms of C on the test instances with 200 jobs.

Instance	C(AMCEA, CH)	C(CH, AMCEA)	C(AMCEA, IMOEA/D)	C(IMOEA/D, AMCEA)	C(AMCEA, TS)	C(TS, AMCEA)	C(AMCEA, ACO)	C(ACO, AMCEA)
I ₂₁₋₁	1	0	0.98	0	1	0	1	0
I ₂₁₋₂	1	0	1	0	1	0	1	0
I ₂₁₋₃	1	0	1	0	1	0	1	0
I ₂₁₋₄	1	0	1	0	1	0	1	0
I ₂₁₋₅	0.86	0	0.95	0.01	0.75	0	1	0
I ₂₂₋₁	0.76	0	0.98	0	0.60	0	1	0
I ₂₂₋₂	0.51	0	0.94	0	0.66	0	1	0
I ₂₂₋₃	0.33	0	0.95	0	0.58	0	1	0
I ₂₂₋₄	0.35	0	0.93	0.01	0.08	0	1	0
I ₂₂₋₅	0.63	0	0.82	0.05	0.75	0	1	0
I ₂₃₋₁	0.30	0	0.99	0	0	0	1	0
I ₂₃₋₂	0.94	0	0	0.50	1	0	1	0
I ₂₃₋₃	0.13	0	0	0.76	0	0	1	0
I ₂₃₋₄	0.63	0	0.75	0.07	0	0	1	0
I ₂₃₋₅	0.57	0	0.88	0	0	0	1	0
I ₂₄₋₁	1	0	0.67	0.30	1	0	1	0
I ₂₄₋₂	1	0	1	0	1	0	1	0
I ₂₄₋₃	1	0	1	0	1	0	1	0
I ₂₄₋₄	1	0	1	0	1	0	1	0
I ₂₄₋₅	1	0	1	0	1	0	1	0
I ₂₅₋₁	0.17	0	0.81	0	0.50	0	1	0
I ₂₅₋₂	0.90	0	0.95	0	0.98	0	1	0
I ₂₅₋₃	1	0	1	0	1	0	1	0
I ₂₅₋₄	0.90	0.03	0.90	0	1	0	1	0
I ₂₅₋₅	1	0	0.86	0.014	1	0	1	0

Table 16
Results of scheduling performance for different electricity pricing schemes.

Pricing scheme	C_{max}	TEC	HV	Peak load reduction rate (stage 1)	Peak load reduction rate (stage 2)
FR1(TOU_{max})	[150–172]	[2395.9–2554.5]	10067.2	68.12	66.92
FR2(TOU_{avg})	[158–217]	[1542.5–1641.7]	58577.9	67.86	66.28
FR3(TOU_{min})	[156–211]	[733.6–769.2]	110858.5	68.12	67.25
TOU	[159–179]	[1522.5–1588.8]	59336.2	68.83	67.95

Table A.1
Beijing’s commercial TOU electricity pricing standard.

Type	High-peak (Jul–Sept)	Peak	Mid-peak	Off-peak
Period	11:00–13:00 20:00–21:00	10:00–15:00 18:00–21:00	7:00–10:00 15:00–18:00 21:00–23:00	23:00–7:00
Price CNY/kWh	1.4409	1.3222	0.8395	0.3818

suggest that, under flat-rate pricing, setting the rate lower than the average value of TOU can help obtain better scheduling performance and more effective peak load reduction.

Beyond the analysis within flat-rate strategies, a direct comparison between the most effective FR3 and TOU yields further managerial insights. By comparing FR3 with TOU, it can be observed that although FR3 yields a higher HV value, its wider C_{max} range suggests a trade-off in production efficiency. Meanwhile, TOU achieves a slightly higher peak load reduction rate, indicating its effectiveness in load shifting. Overall, the dynamic characteristics of TOU pricing provide clearer price signals, which help enhance peak load reduction and contribute to national energy conservation and emission reduction.

6. Conclusions

This paper addresses a bi-objective energy-efficient two-stage hybrid flow shop scheduling problem. The first stage involves parallel machines, while the second involves multiple batch processing machines. The two objectives are to minimize makespan and total energy consumption. To account for factors such as machine qualification, energy consumption rates in different operating states, and batch formation and TOU electricity pricing, a MIP model is formulated, and an enhanced ϵ -constraint method is employed to obtain the exact

Pareto front for small-scale instances using Gurobi. Then, a constructive heuristic algorithm is then proposed to solve medium- and large-scale instances and, and a decomposition-based multi-population bi-objective evolutionary algorithm is developed to improve solution quality further. The parameters of the proposed algorithms are tuned using the Taguchi method. Extensive computational experiments demonstrate the effectiveness of the proposed metaheuristic algorithms in solving the HFS-TOU problem. The proposed algorithms show potential for practical application in manufacturing scheduling scenarios to improve productivity while reducing energy costs.

Future research can be further expanded from problem modeling and solution methodologies. First, on the problem modeling side, more complex and practical scheduling features can be incorporated, such as transportation times between stages, job clusters, machine setup and reset times, and multiple energy consumption modes (e.g., startup energy consumption and maintenance energy consumption), to more comprehensively capture the dynamics of energy use and production in real manufacturing environments. In addition, considering uncertainties such as dynamic job arrivals, machine failures, and maintenance activities can enhance the robustness and applicability of the model. Second, regarding solution methods, adaptive learning-based dynamic control strategies for evolutionary operators can be explored to strengthen the search capability of the bi-objective evolutionary algorithm at

different stages of evolution. Structure-based hierarchical decomposition and region-cooperative search methods also have the potential to accelerate the solving efficiency for large-scale instances. Furthermore, dynamically predicting the distribution characteristics of the Pareto front or identifying promising local search regions could contribute to generating higher-quality solution sets and achieving a better trade-off between energy consumption and productivity.

CRediT authorship contribution statement

Zhao-hong Jia: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis. **Tianfu Wu:** Writing – review & editing, Writing – original draft, Resources. **Han Zhang:** Writing – review & editing, Writing – original draft, Funding acquisition. **Chuang Liu:** Writing – original draft. **Kai Li:** Writing – review & editing.

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Appendix

Table A.2
Orthogonal array for parameter combination pilot experiments.

No.	N_p	$ V $	P_{max}	P_{min}	Loop
1	100	10	0.9	0.1	1
2	100	20	0.8	0.2	2
3	100	30	0.7	0.3	3
4	100	40	0.6	0.4	4
5	100	50	0.5	0.5	5
6	200	10	0.7	0.2	4
7	200	20	0.6	0.3	5
8	200	30	0.5	0.4	1
9	200	40	0.9	0.5	2
10	200	50	0.8	0.1	3
11	300	10	0.5	0.3	2
12	300	20	0.9	0.4	3
13	300	30	0.8	0.5	4
14	300	40	0.7	0.1	5
15	300	50	0.6	0.2	1
16	400	10	0.8	0.4	5
17	400	20	0.7	0.5	1
18	400	30	0.6	0.1	2
19	400	40	0.5	0.2	3
20	400	50	0.9	0.3	4
21	500	10	0.6	0.5	3
22	500	20	0.5	0.1	4
23	500	30	0.9	0.2	5
24	500	40	0.8	0.3	1
25	500	50	0.7	0.4	2

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